INCH-POUND

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8 March 1994
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7 March 1988

MILITARY SPECIFICATION

TRUCK, LIFT, DIESEL ENGINE DRIVEN, PNEUMATIC RUBBER TIRES WITH TOPHANDLER FOR HANDLING CARGO CONTAINERS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. Scope

1.1 Scope. This specification covers the general requirements for a diesel engine driven lift truck with telescoping tophandler for handling cargo containers. The container handling truck (hereinafter called truck) is a front wheel drive, rear wheel steer, pneumatic tired, diesel engine driven, front loading truck with a telescoping container tophandler as specified herein. Unless otherwise specified, the term truck shall be constructed to mean the container handling truck equipped with the telescoping tophandler.

1.2 Classification. Container handler lift truck(s) lifting capacity and the size of containers to be handled shall be defined by the following parameters.

Beneficial comments (recommendations, additions deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Navy Ships Parts Control Center, Post Office Box 2020, Attention: Code 0361, Mechanicsburg, Pennsylvania 17055-0788 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.2.1 Lifting capacity. A one position code used to designate the required minimum truck lifting capacity, excluding the container handling attachment, when lifting single (ISO) or coupled type containers (TRICON or QUADCON). When coupled tpe containers are specified (see 6.2), the truck minimum lifting capacity shall be capable of handling a total of three (3) coupled TRICON (44,800 pounds/ 20,340 kg) or four (4) QUADCON (40,000 pounds/18,160 kg) type containers. Lifting capacity (rated load) is defined as the total or gross weight (container weight plus content weight) at 48 inch (1220 mm) load center and to the full lift height (see 6.2).

TABLE I. Lifting capacity

CAPACITY CODE	GROSS CONTAINER WEIGHT
A	10,000 Pounds each (4,540 kg) (QUADCON)
B	14,900 Pounds each (6,765 kg) (TRICON)
C	22,400 Pounds (10,170 kg) (ISO 1D)
D	52,900 Pounds (24,015 kg) (ISO 1C)
E	56,000 Pounds (25,425 kg) (ISO 1B)
F	67,200 Pounds (30,510 kg) (ISO 1A)
G	72,800 Pounds (33,050 kg)

1.2.2 Container length. A one position code used to designate container length (see 6.2).

TABLE II . Container length (nominal)

LENGTH CODE	LENGTH
A	4 feet and 9.375 inches (1460 mm) (QUADCON)
B	6 feet and 5.5 inches (1970 mm) (TRICON)
C	10 feet (3050 mm) (ISO 1D)
D	20 feet (6100 mm) (ISO 1C)
E	30 feet (9145 mm) (ISO 1B)
G	35 feet (10,670 mm)
H	40 feet (12,195 mm) (ISO 1A)
J	50 feet (15,240 mm)

1.2.3 Container Height. A one position code used to designate container height (see 6.2).

TABLE III. Container height (nominal)

HEIGHT CODE	HEIGHT
A	6'-10" (2080 mm) (QUADCON)
B	8'-0" (2440 mm) (ISO 1A, 1B, 1C, 1D, TRICON)
C	8'-6" (2590 mm) (ISO 1AA, 1BB, 1CC)
D	9'-0" (2745 mm)
E	9'-6" (2895 mm)

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remaining pieces from destructively tested seals shall be

1.2.4 Container Width. A one position code used to designate container width (see 6.2).

TABLE IV. Container width

WIDTH CODE	WIDTH
A	8'-0" (2440 mm) (ISO 1AA thru 1DX, TRICON, QUADCON

1.2.5 Container Stacking. A one position code used to designate number of containers to be stacked (see 6.2).

TABLE V. Container stacking height

STACKING CODE	CONTAINER
A	One high
B	Two high
C	Three high
D	Specify (see 6.2)

2. Applicable Document

2.1 Government Documents.

2.1.1 Specifications, Standards, and Handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these document shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-P-514	Plate, Identification, Instruction and Marking, Blank.
MIL-C-1283	Can, Gasoline, Military, 5 Gallon.
MIL-L-2104	Lubricating Oil, Internal Combustion Engine, Combat/Tactical Service.
MIL-M-3971	Meters, Time Totalizing, Non-Hermetically Sealed, Electrical:General Specification.
MIL-C-52661	Containers, Cargo.
MIL-C-52788	Containers, Refrigerated, 8 feet x 8 feet x 20 feet, Insulated.

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STANDARDS

FEDERAL

FED-STD-H28	Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-130	Identification Marking for U.S. Military Property.
MIL-STD-162	Material Handling Equipment, Preparation for Shipment, Storage, Cyclic Maintenance, Routine Testing and Processing.
MIL-STD-209	Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-461	Requirements for the Control of Electromagnetic Interference Emissions and Susceptibility.
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-1791	Designing For Internal Air Delivery In Fixed Wing Aircraft.

(Copies of military specifications and military standards required by manufacturers in connection with specific acquisition functions should be obtained from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, Pa 19111-5094.)

HANDBOOK

MILITARY

AFSC DH 1-11	Design Handbook, Air Transportability

(Application for copies should be addressed to: ASD/ENESS Wright Patterson AFB OH 45433.)

2.2 Other Publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the

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issue of the nongovernment documents which is current on the date of the solicitation.

ASSOCIATION OF AMERICAN RAILROADS (AAR)

General Rules Governing the Loading of Commodities on Open Top Cars.

Section 1, Part 3 - Recommended Procedures for Conducting Field Impact Tests of Loaded Open Top Cars.

(Applications for copies should be addressed to the Association of American Railroads, 50 F Street, NW, Washington, DC 20001.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI Z26.1 - Safety Glazing Materials for Glazing Motor Vehicles Operating on Land Highways.

(Applications for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME/ANSI B56.1 - Safety Standard for Low Lift and High Lift Trucks.

ASME/ANSI B56.11.3 - Load Handling Symbols for Powered Industrial Trucks.

(Applications for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, N.Y. 10017.)

EUROPEAN TYRE & RIM TECHNICAL ORGANISATION

Tyre and Rim Data Book.

(Application for copies should be addressed to the European Tyre and Rim Technical Organisation, 32, Avenue Brugmann, Brussels, Belgium.)

FEDERATION EUROPEENNE DE LA MANUTENTION (FEM)

FEM Standard 4.001n, Section IV, Industrial Trucks

(Applications for copies should be addressed to F. E. M., 7th Avenue Hoce, 75382, Paris, France.)

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Paragraph 6l MIL-T-l8490 for paid for exposed and spattered portions.

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 668 - Series I Freight Container-Classification, External Dimensions.

(Applications for copies should be addressed to International Organization for Standardization, 3 Rue de Varembe, 1211 Geneva 20, Switzerland.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE HANDBOOK

SAE J20	Coolant System Hoses.
SAE J96	Flashing Warning Lamp for Industrial. Equipment.
SAE J154	Operator Space Envelope Dimensions for Off-Road Machines.
SAE J180	Electrical Charging Systems for Construction and Industrial Machinery.
SAE J185	Access Systems for Off-road Machines.
SAE J198	Windshield Wiper Systems-Trucks, Buses and Multipurpose Vehicles.
SAE J209	Instrument Face Design and Location For Construction and Industrial Equipment.
SAE J297	Operator Controls on Industrial Equipment.
SAE J386	Operator Restraint Systems for Off-Road Work Machines.
SAE J429	Mechanical and Material Requirements for Externally Threaded Fasteners.
SAE J492	Rivets and Riveting.
SAE J514	Hydraulic Tube Fittings
SAE J516	Hydraulic Hose Fittings.
SAE J518	Hydraulic Flanged Tube, Pipe, and Hose Connections, 4-Bolt Split Flange Type.

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terminals of the system, each unit and each assembly (see 5.4.3).

SAE J534	Lubrication Fittings.
SAE J537	Storage Batteries.
SAE J541	Voltage Drop for Starting Motor.
Circuits.	
SAE J551	Performance Levels and Methods of Measurement of Electromagnectic Radiation from Vehicles and Devices (20-1000 MHz).
SAE J553	Circuit Breakers.
SAE J554	Electric Fuses (Cartridge Type).
SAE J598	Sealed Lighting Units for Construction, Industrial and Forest Machinery.
SAE J636	V-Belts and Pulleys.
SAE J680	Location and Operation of Instruments and Controls in Motor Track Cabs.
SAE J759	Lighting Identification Code.
SAE J833	Human Physical Dimensions.
SAE J844	Nonmetalic Air Brake System Tubing.
SAE J845	360 deg Warning Devices for Authorized Emergency, Maintenance, and Service Vehicles.
SAE J898	Control Locations for Off-Road Work Machines.
SAE J899	Operator's Seat Dimensions for Off-road Self-Propelled Work Machines.
SAE J917	Marine Push-Pull Control Cables.
SAE J925	Minimum Access Dimensions for Off-Road Work Machines.
SAE J919	Sound Measurement-Earth Moving Machinery-Operator-Singular Type.
SAE J930	Storage Batteries for Off-Road Work Machines.

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SAE J985	Vision Factors Considerations in Rear View Mirror design.
SAE J994	Alarm- Backup- Electric- Performance, Test, and Application.
SAE J1362	Graphical Symbols for Operator Controls, and Displays On Off-road Self-Propelled Work Machines.
SAE J1402	Automotive Air Brake Hose and Hose Assemblies.
SAE J1703	Motor Vehicle Brake Fluid.

(Applications for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, Pennsylvania 15096).

TIRE AND RIM ASSOCIATION (T&RA)

T&RA Year Book.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 175 Montrose West Avenue, Suite 150, Copley, Ohio 44321).

UNDERWRITERS LABORATORIES (UL)

UL 558 Internal Combustion Engine-Powered Industrial Trucks

(Applications for copies should be addressed to: Underwriter's Laboratories Inc., 333 Pfingsten Road, Northbrook, Illinois 60062).

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the requirements section 3 of this specification shall take precedence. Nothing in this specification, however, shall supercede applicable laws and regulations unless a specific exemption has been obtained.

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3. Requirements.

3.1 First Article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.1, 6.2 and 6.3).

3.2 Environmental Requirements.

3.2.1 Operating Temperature. Unless otherwise specified (see 6.2), the engine shall start within 5 minutes and the truck shall operate as specified herein within 15 minutes after engine start in any ambient temperature from 0°F (-18°C) to plus 115°F (46°C)

3.2.2 Rain. The truck shall start within 5 minutes and operate as specified herein when subjected to a simualted rainfall of not less than 4 inches (100 mm) per hour, (a) without malfunction of any electrical component, gage or instrument; (b) without leakage of water in the air-intake filter; (c) without leakage of water into the torque converter oil chamber, engine crankcase sump, transmission, transfer case, instruments, gages, or fuel tank; and, (d) without not more than one quart (.0009 m^3) leakage into the cab.

3.2.3 Wind. The truck with rated load at maximum lift height shall operate in wind up to 30 miles per hour (48 km/h).

3.2.4 Salt Spray and Blowing Dust. When specified (see 6.2), the truck shall be protected from ocean salt spray and blowing dust as specified herein.

3.3 Safety. Unless otherwise specified (see 6.2), the truck shall conform to the requirements of ANSI/ASME B56.1, UL558 Type "D" and FEM STANDARD 4.001n. All exposed parts subject to high operating temperatures or energized electrically shall be located, insulated, enclosed, or guarded so as to prevent hazards to operating personnel. All moving parts which are of such nature or so located as to be a hazard to operating and maintenance personnel shall be enclosed or guarded. Protective devices shall not impair the operating functions. Walkway coatings, as specified herein, shall be applied whenever personnel are required to stand on the truck to perform repair and maintenance operations.

3.4 Human Factors Engineering. The characteristics of the truck shall provide for operation by personnel ranging from the small man (5%), clothed, through the large man (95%), clothed, in accordance with SAE J833. Controls and displays shall be labeled with universal symbols in accordance with SAE J1362. Controls shall be located in accordance with SAE J898 and SAE J297.

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Instrumentation shall be in accordance with SAE J985.

3.5 Design. The truck(s) shall be designed for connecting to ISO, TRICON, QUADCON, and military containers, lifting the rated load and stacking containers as specified herein (see 6.2). No adjustments or deration to the lifting capacity or stacking height will be permitted due to the the truck performance and requirements specified herein. Design of trucks covered by this specification shall insure safe operation when conducting container operations in and around military depots and sea ports. Truck shall be designed to permit selection and operation of travel, lifting, stering, tilting, sideshifting and container engagement separately, simultaneously or any combinations thereof.

3.5.1 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubricating.

3.5.1.1 Lubrication Fittings. Lubrication fittings shall conform to SAE J534. Fittings shall be located in a protected position and shall be accessible to a hand-operated grease gun with a 10-inch (250 mm) flexible extension. Accessibility to fittings shall be provided and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility.

3.6 Maintainability. Provision shall be made for adjustment, servicing, or replacement of fan belts, fuel pump and components, hydraulic system pump, components, or assemblies where applicable, oil filter and components, starter, alternator, batteries, wearing parts of steering assembly, tires, wheels, lights, and horn. Step surfaces shall be provided on the truck whenever it is necessary to stand on the truck to perform maintenance operations. When hand-access openings are used on the exterior of the truck, the edge of the opening shall be smooth and shall be provided with a removable or hinged cover, whenever it is required. Dimensions of access openings shall be in accordance with SAE J925 for a bare or normal clothed hand. Coolant and lubricants shall be able to be drained into a container without leakage or interference by on any part of the truck. When drains are located over other parts of the truck, integral tubes, pipes, troughs, or other means shall be used to convey the coolant or lubricant from the drain to the container. Drains shall be accessible without the removal or adjustment of accessories or parts other than opening covers. Drainholes with removable drainplugs or covers shall provide for complete drainage of all reservoirs and driveline housings without disconnecting hose(s). The hydraulic system reservoir, service brake system, transmission housing or reservoir, and engine shall be equipped with not less than a 3/4

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inch (20 mm) diameter fill port and with dipsticks or visible level indicator to determine fluid level. As a minimum, dipsticks shall be graduated with two marks indicating full and add levels. Dipsticks shall be accessible without the use of handtools. The truck shall not require scheduled maintenance at intervals of less than 8 hours operation. Reservoir level checks by the operator on a daily basis will not be considered scheduled maintenance.

3.7 Screw Threads. Screw threads shall be in accordance with FED-STD-H28.

3.8 Fungus and Moisture Resistance. The truck's electrical circuitry, including all components and connections except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment in accordance with good commercial practice.

(a) Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.

(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated. When varnish is used, it shall be applied by spray, brush, or a combination of both to give a minimum dry film thickness of 1 mil (0.025 mm) to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.9 Engine. The basic engine shall be a commerical four cycle, water cooled, industrial type diesel as described hereinafter. All other requirements including cooling system, fuel system, ignition system and accessories shall be as specified herein. The engine shall be equivalent in design, power, and size to the engine normally provided by the end item manufacturer in similar commercial products and shall provide sufficient power and torque to meet the performance and operational requirements specified herein. Emergency manual shutdown provisions shall be furnished when recommended by the engine manufacturer. In addition, the engine shall be equipped with automatic low oil pressure and high coolant water temperature shutdown features.

3.9.1 Power and Speed Rating. The power and speed rating of the engine shall be such that operation of the truck, under any of the operational requirements and conditions specified herein, will not require horsepower in excess of the intermittent horsepower rating of the engine for the applicable governed speed as established by the engine manufacturer.

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3.9.2 Engine Accessories.

3.9.2.1 Alternator. A 12 or 24 volt alternator in accordance with SAE J180, shall be provided. The alternator shall be not less than 50-amp capacity and shall provide charge at engine low idle speed. The alternator shall be driven by the gear train, cog belt, single or matched multiple V-belt(s) conforming to SAE J636. The regulator shall be integral with the alternator.

3.9.2.2 Starting System. A device shall be furnished to prevent reenergizing the starter motor after the engine has been started. Trucks shall be provided with an interlock in the starting system or other means to prevent energizing the starter motor, except when the transmission is in the neutral position. The starter shall be heavy duty and shall be in accordance with the engine manufacturer's recommendations.

3.9.2.3 Batteries. The batteries shall be 12 volt, SAE listed, arranged in parallel or series, and shall conform to SAE J537 and SAE J930. The number of batteries shall be as recommended by the engine manufacturer for the truck to operate in environmental conditions specified herein. Unless otherwise specified (see 6.2), the batteries shall be shipped dry and charged. Electolyte shall be furnished in a separate container suitable for marine transport. All terminals shall be accessible for removal, without requiring disassembly of other components.

3.9.2.4 Battery Mounting. Battery mounting shall be under the engine hood or in a weathertight compartment accessible for battery removal without chafing or other damage to the battery. Batteries shall be located so they can be cleaned, serviced, and removed without removing any component except the quick-release battery box cover when a battery box is furnished. Battery supports, holddowns, and areas around the installation which could be affected by dripping or seepage of acids shall be protected with a corrosion-resistant protective coating. The battery shall be mounted is such a manner as not to interfere with access to engine components (accessories). Battery mounting shall provide for complete support over the entire base of the battery and shall be in such a position that the level of the electrolyte is directly visible without removing the battery from its mounting bracket or requiring the use of tools. Battery restraining clamps shall not be designed for or dependent upon special battery shapes and shall be provided to hold the battery in a fixed position. The battery compartment shall have provision for drainage and gas venting. Cover and positioning shall be protected against short circuiting. The battery cable shall be protected by rubber grommets or insulated passages at entry to the battery box.

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3.9.2.5 Battery Cable Terminals. Battery cables shall be in accordance with SAE J541. Positive and negative cable terminals shall be identified. Terminals secured with a plated corrosion resistent bolt and nut shall be provided on the ends of cables connected to the battery to facilitate disconnection and battery removal.

3.9.2.6 Fuel System. The capacity of the fuel tank shall provide for not less than 8 hours continuous operation during the truck operations test specified herein. The tank shall be baffled to prevent surging of fuel. The tank shall be provided with a hand-operated fuel shutoff valve attached directly to the tank. The tank shall be equipped with a self closing type fill and vent filler-cap assembly painted with red or green enamel which shall incorporate a self-closing cap, screw-locking device for the cap, removable strainer, and shall provide for padlocking. The filler-neck opening shall be not less than 2 inches (50 mm) in diameter and shall be located so that the tank can filled from a 5-gallon (0.019 cubic meter) can. There shall be no obstruction above the fuel tank filler neck to prohibit access to the filler pipe with a straight nozzle filler hose. The tank shall accept fuel at a minimum rate of 17 gallons per minute (.0011 cubic meter per second) without splashback or spillage. The fuel tank shall be equipped with a drainplug at the lowest point of the tank. The drainplug shall be removable with handtools without the removal of any other component. The fuel tank shall be protected by position or metal guards from external damage. Adequate supply and return fuel lines shall be provided to insure continuous operation at full throttle and maximum loads including slope ascension. The fuel system shall include fuel pump, fuel strainer/water separator with provisions for draining sediment, water, and fuel filter(s) in accordance with engine manufacturer's recommendations. The fuel strainer or fuel filter(s) shall be between the fuel tank and transfer pump.

3.9.2.7 Air Induction System. A heavy-duty, replaceable dry-type air cleaner shall be furnished. The air cleaner shall be located or designed to prevent water and foreign objects being drawn into the air cleaner when operated on semiprepared surfaces. The air cleaner shall be constructed so as to permit disassembly without tools for servicing and shall be accessible when the engine cowl, hood, or access plate is opened. The air cleaner shall be in accordance with engine manufacturer's recommendations. When blowing dust protection is required (see 6.2), the truck shall be furnished with a dry type, multi-stage air cleaner with a cyclone type pre-cleaner suitable for filtering blowing dust. Either type air induction system shall be equipped with a restriction indicator to indicate when the filter(s) need to be replaced. The restriction indicator may be mounted on the instrument panel or shall be mounted for

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unobstructed visibility by maintenance personnel when the engine cowl, hood, or access plate is opened.

3.9.2.8 Cooling System. The cooling system shall be of the closed-pressure type, incorporating a radiator, radiator pressure cap, hoses, thermostat,fan, and circulating pump. The cooling system shall maintain an air-to-boil temperature of greater than 110°F (43°C). The cooling system fluid be capable of operation in the operating temperature specified herein without damage to the engine or the cooling system. Draincock(s) shall be furnished for complete drainage of the cooling system.

3.9.2.8.1 Fan. The fan shall rotate on antifriction bearings and be driven by a single or matched multiple V-belts conforming to SAE J636. Fan belts take-up shall be at least 3 percent of the belt circumferential length, plus an allowance for replacment of new belts without forcing. The belts shall be clear of all other components within the adjustment range.

3.9.2.8.2 Thermostat. The thermostat shall be in accordance with the engine manufacturer's recommendations. The thermostats shall be removable by the use of handtools.

3.9.2.8.3 Radiator. The radiator assembly shall consist of a fin and tube-type core with and expansion tank and sediment tank. Where the top of the radiator is lower than the top outlet of the engine, a surge tank may be used. The radiator pressure cap shall be assessible without physical removal of the engine hood. The radiator shall be mounted to prevent its being damaged due to shock and racking experienced in normal operation of the truck. The radiator shall be protected by guards, position or the truck counterweight. A shield shall be provided, if necessary, to protect the radiator from mud and gravel thrown by the steering tires. Hoses shall clear the truck structure and other components under all operating conditions specified herein. The radiator shall include baffles and shroud to prevent reverse airflow back into the engine compartment. When blowing dust protection is required (see 6.2), the truck shall be furnished with a radiator designed for operation in a humid and dusty environment. Radiator shall provide sufficient heat dissipation and designed such that blowing dust will not clog radiator air passages.

3.9.2.8.4 Engine Coolant Hose. All engine coolant hoses shall be in accordance with SAE J20.

3.9.2.9 Governor. A speed-limiting type governor shall be furnished. The engine speed shall not exceed the engine manufacturer's recommended intermittent duty speed.

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3.9.2.10 Exhaust System. The exhaust system shall be protected against entry of rain and shall include drainage provisions to prevent accumulation of water and condensed vapors. The back pressure of the exhaust system, including muffler, shall not exceed the maximum recommended by the engine manufacturer when measured approximately 2 inches (50 mm) beyond the flanged header outlet at any load up to rated net continuous load. The exhaust header outlet shall be provided with a flanged connection for attachment to exhaust system. The muffler shall be independently supported around the body or muffler outlet. If the tailpipe is independent of the muffler or extends more than 12 inches (305 mm) from the muffler, it shall be supported. Exhaust gases shall be emitted vertically above the operator's head or at the rear or sides of the truck within the plan outline of the truck. The exhaust system shall be arranged to prevent burns to the operator when operating the truck and to maintenance personnel while working on the truck.

3.9.2.11 Oil Filters. Oil filter(s) of the replaceaable and full-flow type shall be furnished in accordance with the engine manufacturer's recommendations. The oil filter(s) shall be mounted in an accessible location for ease of replacement of elements. Oil lines (if applicable) to and from the filters shall be installed to minimize vibration and shall be flexible hose.

3.10 Drive Assembly. The drive assembly shall consist of all components necessary to transmit power from the engine to the wheels. All rotating shafts and axles shall be supported on antifriction bearings. All gears shall operate in lubricant, and a standard pressure grease lubrication system or permanent lubrication shall be provided for all friction parts not so lubricated. Gear housing shall be equipped with filler plugs and drain-plugs. Differential ring and pinion gears shall be adjustable. The drive shaft shall be provided with a minimum of two heavy-duty or one double universal joint and a nonbinding slip joint. Drivetrain components shall conform to and shall be installed in accordance with each component manufacturer's recommendation.

3.10.1 Transmission. The transmission shall be the full-torque, powershift type in combination with a torque converter. The transmission shall provide not less than two forward and two reverse gear ratios, of which not less than one shall be manually selected. Transmission speeds and diesel engine operation shall be coordinated such that the truck shall meet all performance and operation requirements specified herein. All transmission gears shall be constant mesh. Shifting betweeen gear ratios shall be accomplished by selective engagement of multiple-disk clutches which shall be controlled by a hydraulic-control valve

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body mounted on or in the transmission housing. The transmission shall provide for shifting under full engine power through all successive gear ratios in each direction. The transmission shall provide controlled application of the clutches to provide smooth shifting without slippage that could damage the clutches. Oil pressure for actuation of the transmission clutches, for torque converter supply, and for transmission lubrication shall be supplied by a pump mechanically driven by the engine. The transmission and torque converter fluid shall be filtered by full-flow filter(s) having replaceable elements. The torque converter and transmission shall absorb maximum stall operation for a minimum of 15 seconds without damage or permanent deformation. The transmission shall provide for positive inching control or declutch control of the truck (throughout the entire range of engine speed) in the forward and reverse directions. The inching or declutch control shall permit tilting and full lifting speeds with rated load while the transmission control is in forward or reverse position without vehicle movement. Transmission shall be furnished with oil dipstick inserted into the transmission reservior. Dipstick shall indicate "add" and "full" levels. An oil cooler shall be provided when necessary to meet the environmental and performance requiements specified herein

3.11 Hydraulic System. The hydraulic system shall consist of all hydraulic components necessary for operation of the mast, tophandler, steering system, and forks, and shall not include the brake and transmission systems. The system shall have a maximum working pressure (relief valve setting) not in excess of 3000 psi (20,684 Kpa). Oil for the system shall conform to MIL-L-2104, Grade 10. All hydraulic component materials shall be compatible with these oils. The temperature rise of the hydraulic fluid shall not exceed 115°F (46°C) above ambient when the truck is in operation during the first article inspection. All system components which are subject to full system pressure shall withstand a proof pressure that is equal to or greater than 1-1/2 times the maximum working pressure (1-1/2 times relief valve setting) without external leakage, damage or pressure deformation. All hydraulic system components that are subject to full system pressure shall have a minimum burst pressure that is equal to or greater than 2 times the system relief valve setting. Hydraulic fittings on pumps, valves and cylinders shall be O-ring straight-thread or pipe thread type as specified in SAE J514 for ports 1 inch (25 mm) and smaller, and 4-bolt flanged type as specified in SAE J518 for ports larger than 1-1/4 inch (32 mm). The hydraulic system shall be cleaned free of foreign matter including weld slag and spatter. Hydraulic connection ports of the hydraulic system component shall be sealed by the component manufacturer. The system shall provide for lowering of rated

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In accordance with the appendix of the box specification.

load at the maximum rate specified herein in the event of failure of or damage to hydraulic hose supplying the lift cylinder.

3.11.1 Hydraulic Lines. Hydraulic lines shall consist of tubing, hose and fittings. Hose or a combination of hose and tubing shall be used between components that are not mounted to a nonflexing, rigidly connected structural member or rigidly connected subassembly. All lines shall be routed to provide the minimum number of bends, ease of maintenance, and maximum protection. Provisions shall be made to prevent damage of lines due to chafing. The hose installation , such as angle of fittings and location, shall cause no stress concentration on the hose at the fitting, and the hose shall be supported, when necessary, to eliminate sagging at the fitting. All lines shall be cleaned free of foreign matter including weld slag and spatter.

3.11.2 Quick Disconnect Fittings. Quick disconnect fittings shall be furnished for all hydraulic lines which connect the tophandler to the truck. Quick disconnect couplings shall connect to lines smaller than 1 inch (25 mm) with an o-ring boss conforming to SAE J514. The couplings shall connect to lines 1-1/4 inch (32 mm) and larger with SAE J518, 4-bolt, split-flange face, 3,000 psi (20,684 KPa) . The coupling halves shall be restrained to prevent contact with the ground or any surface of the hoist or tophandler when connected or disconnected. The coupling halves shall be identified with metal identification plates, metal tags, or metal or plastic bands to assure assembly of correct mating halves. Each coupling half shall be provided with a captive chained cap to protect the opening when the couplings are disconnected or permanently affixed receptacles with caps to protect the opening when the coupling halves shall mate together when not in use to prevent corrosion, damage, and contamination of mating surfaces. Couplings shall not have any partially coupled unlocked positions in which the couplings can remain stable and permit flow. When either coupled or uncoupled and with 60 psi (415 KPa) applied, the couplings shall seal the ends of the disconnected lines without permitting a fluid loss exceeding 1/2 fluid ounce (.000015 cubic meter). At full line flow, maximum pressure drop across quick disconnect couplings shall be 15 psi (103 KPa). The connected couplings shall withstand 3,000 psi (20,684 KPa) without evidence of leakage or rupture.

3.11.3. Hydraulic Pump(s). The hydraulic pump(s) shall withstand a minimum proof pressure of 1-1/2 times the maximum system working pressure without permanent deformation, damage, or external leakage.

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3.11.4. System Filtration. System filter(s) shall be furnished either in the suction line, pressure line, return line or resevoir. The filter shall have a proof pressure equalling 150 percent of system maximum working pressure. When blowing dust protection is required (see 6.2), the truck shall also be furnished with a minimum of two 10 micron (nominal) filters.

3.11.5. Reservoir. A reservoir shall be furnished to contain the hydraulic system oil. The reservoir shall include a filler opening, screen, cap, and other items required to function as specified herein. The reservoir shall be mounted to prevent vibrations that could result in leaks, loosening of fasteners, brinelling of material, or cracks. The location shall not impair operational visibility or access to either side of the operator's compartment. The reservoir shall have sufficient capacity to prevent air entering the system with all hydraulic pistons fully extended and shall have sufficient free air capacity to prevent oil from being discharged through the reservoir air vent when maximum return flow of oil is surged into the reservoir from the system. The add and full marks on the dipstick or sight-glass shall indicate the correct oil level when the truck is on level ground, the mast is vertical, and the tophandler is at lowest level and the oil is at normal operating temperature. The filler cap and tube shall be located to provide for filling from a standard 5-gallon (0.019 cubic meter) container conforming to MIL-C-1283, and to prevent entrance of contaminants throughout filling and checking operations. The reservoir filler neck shall contain a removable strainer retained in place unless all oil added to the reservoir is filtered by a finer filter prior to passing through the pump. The filler tube shall be not less than 1 1/2 inch (40 mm) inside diameter. The reservoir pump suction openings shall be located a minimum of 3/8 inch (10 mm) above the bottom of the reservoir. Except for sealed reservoirs a two way breather cap with filter shall be provided and shall filter incoming air. When blowing dust protection is required (see 6.2), the filter rating shall be 50 micron (nominal). The breather shall preclude the entrance of water induced by rain. The reservoir shall provide a tortuous flow path from the reservoir inlet to the reservoir outlet in order to prevent a direct current between the two ports. All return line flow except the vent line shall discharge into the reservoir at a level which is below the oil level in the reservoir when the lift cylinder is at maximum extension.

3.11.6. Relief Valve. A hydraulic-system valve shall be provided to protect the hydraulic system from excessive pressure. The valve shall have a reseat pressure of not less than 85 percent of rated flow pressure settings at an oil temperature of 150°F (66°C) plus or minus 10°F (-12° C). The response of the

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valve shall be such that no pressure peaks shall exceed 150 percent of the relief pressure.

3.11.7. Hydraulic Control Valves. Mounting of the valves shall be rigid and shall permit no vibration which may cause instability of the controls. All control-valve spools shall be spring centered with sufficient force to return the levers to the center position and positively retain them there throughout all operations. Spools shall not bind under all operating conditions specified herein. All exposed control-valve surfaces shall show no evidence of corrosion when tested as specified herein.

3.11.8. Cylinders. All hydraulic system and steering cylinders shall be fabricated from either seemless steel tubing, or welded and redrawn tubing. All welding and brazing to the cylinder within the zone of the piston operation shall be done prior to final machine honing or rolling of the cylinder. Cylinders shall be so located to provide for ease of maintenance, and replacement. All internal cylinder ports shall be located beyond the area of piston travel. Each cylinder port shall have an exclusion device to prevent foreign material and fluids from entering around the piston rod and damaging the seal and other cylinder components. All hydraulic system and steering cylinders shall perform as specified herein without buckling or bending. Cylinder rods shall be hard chrome plated. After plating, maximum rod surface roughness shall not exceed 32 microinches (0.008 mm). When blowing dust protection is required (see 6.2), the cylinders shall first receive a nickle chrome plating and then finished with hard chrome plating thin buffed. All cylinders shall withstand a pressure that is equal to 1.5 times the system relief valve setting without evidence of rupture, permanent deformation, damage, or external leakage. Lift cylinders, if single-acting type, shall provide a bleeder line to reservoir or intenal valve to return oil that bypasses the piston.

3.11.9. Hose Reels. When hydraulic hose reels are required on the mast to maintain hose tension during mast extension, sheave wheels, guide rollers, guide blocks, or other means shall be provided to prevent the hose from becoming entangled around and riding over the reels or reel flanges.

3.11.10. Dust Protection. When blowing dust protection is required (see 6.2), the truck shall be furnished with the following protection:

3.11.10.1. Wiper/Scraper. All mast cylinders, top handler extension and twist lock cylinders, and fork positioner cylinders shall be equipped with wiper/scrapers to prevent dust from entering the hydraulic fluid system and to prevent premature

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failure of cylinder rod surfaces and packing due to blowing dust. Wiper/scrapers shall be a twin seal arrangement in a modified cylinder head bushing.

3.11.10.2. Guards. All tophandler slew and tilt cylinders, sideshift cylinders, and steering cylinders rods shall be protected throughout their full stoke by an enclosing guard or expandable bellows. Guard shall provide protection from direct impingement of blowing dust and to prevent dust from entering the hydraulic, brake and steering fluid systems. The guards or bellows shall be constructed of moisture resistant materials.

3.12 Cold weather system aids. The truck shall be equipped with cold weather system aids necessary to meet the environmental requirements specified herein. Aids shall include battery blankets, glow plugs or fluid priming engine starting aids, engine oil heater, hydraulic system heater, engine coolant heater, and transmission heater. However, only the aids necessary to meet the temperature requirements specified herein will be required. Heaters shall be electric and shall operate on 110 volt alternating, 60 cycle electrical current. When a fluid priming starting system is provided the size of the primer shall be in accordance with the engine manufacturers recommendations. The primer shall be mechanically or electromechanically operated. When the primer is mechanically actuated, it shall be furnished with a total cable length not exceeding 6 feet (1825 mm) and a "tee" handle and when electromechanically actuated, it shall be furnished with a temperature control.

3.13. Uprights, Carriage, Tophandler Assembly and Fork Assembly. Uprights, when furnished, shall be of the telescopic, roller type. Rollers shall be of the permanently-lubricated-for-life type or equipped with lubrication fittings. Uprights shall be positioned is such a manner that the back surface of the lower fork mounting plate or crossbars in the vicinity of the tires are located forward of the vertical plane tangent with the front face of the drive tires. The drive tires shall not interfere with the carriage assembly travel at any upright tilt angle. The lift cylinder shall be mounted and positioned so that it does not protrude forward of the plan outline of the uprights. Uprights and carriage assembly shall be removable from the truck without requiring permanent modification (such as cutting or welding) to the truck. Crossmembers of the mast structure, when in a lowered position, shall not interfere with the operator's horizontal or downward line of sight to the front of the truck. Adequate means shall be provided in the uprights to compensate by replacement or adjustment for wear which may develop between lateral rolling contacts. Positive means shall be provided to prevent overtravel of the carriage or channel in both upper and lower positions. Carriages shall be supplied with adjustable thrust bearings to

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resist thrust introduced by off-center loading. Chains and chain anchors used in the lifting mechanism shall have a factor of safety of not less than 4.6 to 1 based on minimum ultimate tensile strength when truck is carrying rated load equally distributed on the chains. The upright shall be provided with a positive means of preventing the lowest part of the carriage or gallows from being lowered to a point less than six inches (150 mm) above ground level.

3.13.1. Cantilever Boom. When a cantilever boom is furnished in lieu of the uprights specified in 3.13 means shall be provided to maintain load to within 2 degrees of the same relative angle with the vehicle frame when traversing the entire range of lift operation. Means shall be provided to prevent cavitation or excess hydraulic pressure in the hydraulic lift system for any combination of lift and tilt operation specified herein; and when activation lift or tilt control with engine off.

3.13.2. Telescoping Container Tophandler. Unless otherwise specified (see 6.2), the truck shall be furnished with a hydraulically operated telescoping (expandable) container tophandler to handle ISO 20, 30, and 40 foot (6,095, 9,145, and 12,190 mm) long cargo containers. When specified (see 6.2), an expandable type tophandler to handle one, two and three TRICON or one, two, three and four QUADCON type containers shall be furnished in lieu of handling ISO containers. The tophandler shall be compatible with containers 1A through 1CX referenced in ISO 668, MIL-C-52661, MIL-C-52788. The tophandler shall be an expandable, single horizontal unit that will safely mate, align, couple, and lock with the containers specified. The tophandler shall safely lift, transport, and lower the coupled containers at specified rates of hoist and lower velocities. The tophandler shall mount to the truck by positive connection to the carriage or boom. Hydraulic lines and electrical lines between truck and tophandler shall be furnished with quick-disconnects. A single lifting padeye shall be mounted on each corner of the tophandler. Padeyes must be mounted within twelve inches of the twistlock. Padeyes shall be mounted in such a position as to allow for easy chain or cable attachment to halfrack containers or other loads without interfering with normal container handling. Each padeye shall be designed and built to a safe working load of ten tons and shall be equipped with equally rated shackles.

3.13.2.1 Coupling Devices. Coupling devices shall be furnished to mate the tophandler to ISO 668, SIXCON, QUADCON, MIL-C-52661, and MIL-C-52788 containers. Coupling devices shall be hydraulically operated. One coupling device shall be provided for each corner of the container. The top lifting coupling device shall be of the "universal" twist-lock type. The location of the coupling devices shall coincide with the longitudinal,

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lateral, and diagonal dimensions and tolerances for the container corner fittings in accordance with ISO 668, SIXCON, QUADCON, MIL-C-52661, MIL-C-52788 specifications. Hooks, shackles, and clevis attachments for direct mating of the truck with the container corner fittings are not acceptable in lieu of the twistlocks, but may be furnished as redundant coupling devices.

3.13.2.2 Aligning Arms. The tophandler shall be equipped with a minimum of two aligning arms located on the longitudinal frame rail nearest the mast assembly. The minimum length below the rail shall be 18 inches (455 mm). The end of the arms shall be flared for a minimum of 3 inches (75 mm). Alignment arms shall be located at each end of the frame rail to coincide with the corner fittings of the containers.

3.13.2.3. Out of level container, Slew and Reach. Tophandler shall be furnished with system(s) which permit the tophandler to slew, reach and connnect to out of level container.

3.13.3. Fixed Container Tophandler. When specified (see 6.2), the truck shall be furnished with a fixed type container tophandler to handle one 10, 20, 30, or 40 foot (3050, 6,100, 9,145, and 12,190 mm) long ISO or military cargo container (see 6.2). The tophandler shall be a single horizontal unit that will safely mate, align, couple, and lock with one specified container. The tophandler shall safely lift, transport, and lower the container at specified rates of hoist and lower velocities. The tophandler shall mount to the truck by positive connection to the carriage or boom. Hydraulic lines and electrical lines between truck and tophandler shall be furnished with quick-disconnects. A single lifting padeye shall be mounted on each corner of the tophandler. Padeyes must be mounted within twelve inches of the twistlock. Padeyes shall be mounted in such a position as to allow for easy chain or cable attachment to halfrack containers or other loads without interfering with normal container handling. Each padeye shall be designed and built to a safe working load of ten tons and shall be equipped with equally rated shackles.

3.13.3.1 Coupling Devices. Coupling devices shall be furnished to mate the tophandler to ISO 668, MIL-C-52661, and MIL-C-52788 containers. Coupling devices shall be hydraulically operated. One coupling device shall be provided for each corner of the container. The top lifting coupling device shall be of the "universal" twist-lock type. The location of the coupling devices shall coincide with the longitudinal, lateral, and diagonal dimensions and tolerances for the container corner fittings in accordance with ISO 668, MIL-C-52661, and MIL-C-52788 specifications. Hooks, shackles, and clevis attachments for direct mating of the truck with the container corner fittings are

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not acceptable in lieu of the twistlocks, but may be furnished as redundant coupling devices.

3.13.3.2 Aligning Arms. The tophandler shall be equipped with a minimum of two aligning arms located on the longitudinal frame rail nearest the mast assembly. The minimum length below the rail shall be 18 inches (455 mm). The end of the arms shall be flared for a minimum of 3 inches (75 mm). Alignment arms shall be located at each end of the frame rail to coincide with the corner fittings of the containers.

3.13.3.3. Out of level container, Slew and Reach. Tophandler shall be furnished with system(s) which permit the tophandler to slew, reach and connnect to out of level container.

3.13.4 Forks. Unless otherwise specified (see 6.2), truck shall also be equipped with forks, fork positioner and sideshift to permit conventional forklift operation. Forks shall be capable of withstanding, without permanent deformation, 150 percent of rated load plus 50 percent of the tophandler weight at 48-inch (1220 mm) load center measured from face of the forks. Forks shall be attached to the fork carriage in a manner which prevents lateral slippage or permanent deformation of fork carriage components. Forks shall be a maximum of 5-3/4 inches (145 mm) thick at the heel, a maximum of 14 inches (355 mm) wide, and 96 plus 0 inch minus 1/2 inch (2440 mm plus 0 mm minus 15 mm) long. Fork thickness shall not exceed 1-1/4 inch (30 mm) at the tips and the end taper shall not be less than 36 inches long (915 mm). Taper shall be on the underside of the forks. The top of the fork tips shall be chamfered or rounded to break corners and provide smooth edges. Each fork arm shall be stamped with its individual rated capacity in a visible area not subject to wear.

3.13.4.1 Fork Positioner. Hydraulic powered fork positioners shall be furnished. Fork spacing shall be simultaneously adjustable.

3.13.4.2 Fork Sideshift. Truck shall be equipped with hydraulically powered sideshift.

3.14. Wheels and Tires.

3.14.1. Wheels. The wheels shall be fastened to the hubs in accordance with the axle manufacturer's standard for off-road applications. Wheel rims shall be listed in the Tire and Rim Association (T&RA) Yearbook or Tyre & Rim Data Book. Spacing between rims of dual tires shall be such as to prevent contact of the tires when the truck is on a level surface with rated load in a static condition.

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3.14.2. Tires. The tires shall be new and unused. The tires shall be the pneumatic, high pressure, tube or tubeless type furnished with non-directional tread. Tires, valves and caps sizes shall be listed in either the T&RA Yearbook or European Tyre & Rim Technical organisation. Extended tube or rim valve stems shall be provided for each inner tire which will permit inflation of the inner tire beyond the face of the outer wheel.

3.15. Steering. Power steering shall be furnished. An automotive type steering wheel mounted on the steering column, shall turn the steer wheels. There shall be a minimum of 7 inches (180 mm) measured vertically between the lowest point on the steering wheel rim and a line parallel to the floor and passing through the highest point of the unoccupied seat cushion. There shall be a minimum of 3 inches (75 mm) between the rim of the steering wheel and the directional control lever(s) for all positions of the lever(s). Clockwise rotation of the steering wheel shall provide for right-hand turning of the truck while in forward motion. Provisions shall be made for precision adjustment for wear and alignment of all major steering components and for lubrication of all friction points by accessible grease fittings or permanent lubrication, or a lubricant reservoir. The reservoir shall be accessible for filling and checking of the lubricant level. Means shall be provided to prevent wheels or tires or steering linkage from contacting any part of truck structure, other than stops, irrespective of obstacle magnitude or cramped steer angle.

3.16. Brake System. The truck shall be equipped with a powered service brake system capable of safely stopping, holding, and controlling the movement of the truck under the operating conditions and in the environments specified herein. The service brake system shall include all components required for a complete matched brake system based on the truck gross weight (truck weight plus rated load). The characteristics and installation shall be in accordance with the recommendations of the brake assembly manufacturer. Brake lining for shoes or pads shall be intended for performance requirements specified herein and shall be securely bonded or fastened to the brake components. Brake lining shall have at least 80 percent of the total lining rubbing surface in contact with the wearing surface during the service life of the lining. Automatic or mechanical brake adjustment shall be provided. Automatic adjustment shall maintain correct lining to drum or disc adjustment without looseness or lockup. The automatic adjusters shall be of the guarded or enclosed type, protected from environmental damage or malfunction. The necessary provisions to permit manual brake adjustment shall be provided.

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3.16.1. Service Brake System. The service brake system shall be powered and shall be either hydraulic, air or air-over-hydraulic type. The power brake system shall be independent of the hydraulic and drive assembly systems. Brake fluid shall be in accordance with SAE J1703. The power source shall charge the system to maximum power setting within 2 minutes when the brake actuating energy is fully exhausted. With no further addition of brake system actuation energy, a sufficient energy reserve shall be furnished such that a fully charged system shall have the capacity of three full effective service brake applications. The brake system shall provide braking, free of brake chatter and free of feed-back amounting to more than 1/4 inch (6 mm) total movement measured at the foot pedal when actuated at all speeds. An audible and visual low-pressure warning device shall be furnished as specified herein.

3.16.2. Straight Airbrake System. When a straight airbrake system is furnished, the system shall include not less than a 12 cfm (0.0057 cubic meter per second) air compressor, alcohol evaporator or automatic moisture ejectors/filters, and wet and dry reservoirs. The air compressor shall be engine-driven, engine-lubricated and water-cooled. Previously cleaned air for the compressor shall be piped from the engine air cleaner intake, and check valves shall be provided to prevent reverse airflow from the compressor and the evaporator. All reservoirs shall be furnished with automatic or remote control drain valves. All lines (except drains) connected to the reservoirs shall be attached on or above the horizontal centerline of the reservoirs. The airbrake plumbing shall be routed, supported and fastened to provide maximum protection. In-cab plumbing and chassis plumbing not subject to road hazards shall conform to SAE J844, Type 3. Chassis plumbing subject to road hazards shall conform to SAE J1402, Type D, Class II, flexible hose with SAE 100R5 reusable fittings as specified in SAE J516. The air compressor discharge line shall be teflon with copper tubing or teflon with stainless steel braiding.

3.16.3. Air-Over-Hydraulic Brake System. When an air-over hydraulic brake system is furnished, the complete air system shall also be in accordance with 3.16.2.

3.16.4. Straight hydraulic brake system. When a straight hydraulic sytem is furnished, the system shall utilize a master cylinder. Wheel brake cylinders shall be located such that hydraulic lines may be bled of air without the necessity of removing the wheels or disconnecting hydraulic lines. When the hydraulic master cylinder is located under the floor plate, a access opening shall be provided thru a hole that will accommodate a 2 1/2" (65 mm) diameter filler line with its center being concentric with the center of the fill port.

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3.16.5. Parking Brake System. The truck shall be equipped with a parking brake. The parking brake shall be an independent mechanical friction brake mounted on the drive or propeller shaft rotating with the wheels or an independent mechanical system which actuates the service brakes. The parking brake(s) shall be hand or foot operated and shall be equipped with a locking device. The manual control shall be located within reach of the operator and in a noninterfering position, permitting easy and safe movement on and off the vehicle. The effort required to set the parking brake lever shall not exceed 90 pounds (40 kg) measured from the extreme end of the operating lever. If the land lever is of the ratchet type, the ratchet and pawl shall be of heat-treated steel. Adjustment of parking brakeshoes or bands shall be accomplished without the removal of any major assembly other than tires and wheels. The parking brake may be hand operated in accordance with SAE J680.

3.17. Truck Access. Operator access systems to the truck and operator's compartment shall be in accordance with minimum requirements of SAE J185.

3.18. Seat. A cushion seat and backrest shall be furnished in accordance with the minimum requirements specified in SAE J899. Horizontal seat adjustment shall be by the operator from the operating position without the use of tools. The seat cushion and backrest shall be of foam rubber or polyurethane covered with slip coated vinyl upholstery having a minimum weight of 40 ounces (1.1 kg)per yard (915 mm) for 54-inch (1370 mm) width. The seat location shall conform to SAE J898.

3.18.1. Seatbelts. Seatbelts shall be furnished in accordance with SAE J386,Type I.

3.19. Cowl or Hood. The engine shall be completely enclosed with a metal hood cover and side plates. Either the hood cover or side plates shall be quick opening. The hood cover, if lift-up type, shall be held in the open position in a safe manner by counterweight or spring and linkage mechanism or latch, or brace which cannot be inadvertently released while servicing the truck. The hood or side plates shall be equipped with a mechanical hold-open device which will lock the hood cover or side plates in the open position. Hood cover shall be capable of being opened or removed by one man. Handtools shall not be required for release of hood or side plates. If finger ring lift latches are provided, the force required to release or fasten the latches shall not exceed 15 pounds (7 kg). All engine components (accessories) shall be completely accessible, without requiring the removal of any additional truck structure, when hood cover and side plates are opened. Fiberglass construction is not acceptable for cowl or hood.

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3.20. Wheel Guards. Wheel guards shall be provided to prevent foreign objects being discharged in the direction of the operator, endangering personnel or otherwise creating a hazardous situation.

3.21. Towing Device. A pintle or pin-type towing device shall be provided in the center rear of the truck. The towing device shall not protrude beyond the path followed by the rear end of the truck when making a turn of minimum turning radius.

3.22. Slinging and Tiedown Provisions. Permanently affixed slinging and tiedown provisions that enable the truck, mast, and tophandler to be lifted in its normal travel position and to be fastened to the floor or deck of a transportation medium shall be provided. Slinging and tiedown attachments shall conform to MIL-STD-209, Class 1, 2 or 3 for type II or III equipment and shall also conform to AFSC Design Handbook DH 1-11 when air transportability is required (see 6.2). When required, due to truck size and weight, slinging and tiedown sizes may be increased above the sizes indicated in MIL-STD-209. However, increased slinging and tiedowns must still meet the strength requirements of MIL-STD-209 and AFSC Design Handbook DH 1-11.

3.23 Instrument panel. The instrument panel shall be located so that the operator can view the instruments from his normal operatin position. The instrument panel shall be installed on an inclined or vertical plane. Panel design and instrument grouping shall be in accordance with SAE J209. On this panel shall be mounted the voltmeter or ammeter, engine oil pressure gage or indicator light, engine coolant temperature gage or indicator light, transmission fluid temperature gage or indicator light, fuel gage, engine run control, starter switch, airbrake pressure gage, and any other instruments when used. Whenever indicator light(s) are furnished, means shall be provided to check operation of the light(s). Instrumentation shall be moisture and weather resistant. All gages, indicator lights, and switches shall be identified on the face of the gage as to nomenclature of component and fluid being monitored and positions and function control. Symbols used on the panel shall be in accordance with SAE J1362. All gages shall be mounted flush and internally lighted and shall be controlled by the forward floodlight switch.

3.23.1. Electrical System Monitor. A remote indicating ammeter or voltmeter or alternator indicator light (red or amber) shall be furnished. The ammeter shall be of sufficient capacity to indicate variations in current supply or demand experienced by the truck, exclusive of starter requirements. The indicator light shall light when the alternator is not charging and the engine run control is "on".

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3.23.2. Engine Oil Pressure Monitor. A remote indicating engine oil pressure gage or a low-pressure red or amber indicator light shall be furnished.

3.23.3. Engine Coolant Temperature Monitor. A remote indicating engine coolant temperature gage or a high-temperature red or amber indicator light shall be furnished. The light shall come "on" when the engine coolant temperature exceeds the temperature specified by the engine manufacturer.

3.23.4. Transmission Temperature Monitor. A remote indicating transmission temperature gage or a high temperature amber or red indicator light shall be furnished. The light shall come "on" when the transmission temperature exceeds 250°F (120°C).

3.23.5. Hour Meter. An electrically operated hour meter conforming to MIL-M-3971 Type I and MIL-M-3971/1, which registers the number of engine operating hours only, shall be provided. The hour meter shall be accessible for reading without removal of any component of the truck.

3.23.6. Fuel Gage. An electric, thermally stabilized, remote indicating or mechanical fuel gage shall be provided.

3.23.7. Air Pressure Monitor. When a airbrake or an air-over-hydraulic brake system is furnished, a remote indicating brake air pressure gauge or a low-pressure red or amber indicator light and audible alarm shall be furnished. Low-pressure shall energize the audible alarm. Silencing switch is not allowed.

3.23.8 Transmission Pressure Monitor. A remote indicating transmission pressure gauge or a low-pressure red or amber indicator light shall be furnished.

3.24. Controls. Each control shall be provided with a position marking diagram on a plate or embossed, or depressed on or near the control. Controls shall conform to ASME/ANSI B56.1 and ASME/ANSI B56.11.3. The diagrams shall be visible to the operator from his normal operating position. Letter height for control markings shall be equal to or greater than that specified for general dial and panel design for the applicable viewing distance of the 95th percentile man, with seat adjusted to rear position, as specified in SAE J833. All controls for the truck shall be located in the operator's compartment. The distance from the near edge of one control to the near edge of an adjacent control or other surface shall be a minimum of 1-1/2 inches (40 mm). When controls are toggle type intended for finger operation, separation of the controls shall be not less than 1/2 inch (15 mm). Controls shall be moisture and weather resistant.

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Mechanical push-pull remote controls, when furnished, shall be of the waterproof antifriction type, conforming to requirements for marine cables specified in SAE J917. Bend radii shall be as recommended by the cable manufacturer. The manufacturer's published ratings for continous operation shall not be exceeded for any required operation of the control specified herein. Forces necessary to actuate directional and hydraulic controls shall be 10 pounds (5 kg) maximum. All hand-contol levers and linkages shall withstand a minimum force of 50 pounds (25 kg) applied at the handgrip in the normal directions of operation and shall have sufficient strength to withstand, without permanent deformation or damage, all vibration forces generated during all operations specified herein. The travel of the controls shall not exceed 6 inches (150 mm) from the center postion when measured at the end of the grip.

3.24.1. Light Switches. A minimum of three light switches shall be furnished on the instrument panel to control the floodlights specified herein. One switch shall control the forward floodlight(s), one switch shall control the rear floodlight(s) and one switch shall control the twistlock floodlights.

3.24.2. Engine Run Control. An engine run control shall be furnished and shall be a key type. It shall have not less than two positions "on" and "off". The diesel engine operation shall be terminated when tne engine run control is positioned to "off". The switch shall control the operation of monitors and of all electrical equipment except the horn. Whenever a mechanical engine shutoff and electrical fuel pump are furnished a warning buzzer shall be furnished to sound when key is "on" and engine is not running. Two keys shall be furnished with each truck. All keys shall be identical for each truck delivered on a single contract.

3.24.3. Starting-Motor Switch. The starting motor shall be energized through an auxiliary solenoid and a starter solenoid connecting the starting motor to the battery. The solenoids shall be operated by depressing a button installed on the instrument panel or turning a spring loaded switch to a "start" position. The starter shall be inoperable when the engine run control is in the "off" position and the engine is running.

3.24.4. Braking and Inching or Declutch Control. One or two pedals or treadles shall be provided for inching or declutch and for braking control. A right pedal or treadle located for convenient right foot operation shall be furnished and may provide braking control only or may be a combination inching-braking control. A left pedal or treadle located for convenient left foot operation shall provide inching or a combination inching-braking control. Each pedal or treadle shall

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be a minimum of 3 inches (75 mm) wide.

3.24.5. Accelerator Control. The accelerator control shall be installed for comfortable right-foot operation and shall be located to the right of, and shall be free of interference from, the brake pedal. The actuation force on the accelerator measured at the top shall be a minimum of 10 pounds (5 kg) and a maximum of 20 pounds (10 kg) through the total travel of the accelerator with the engine running.

3.24.6. Transmission Controls. Selective forward and reverse and speed range controls shall be provided and shall be located for convenient hand operation. Movement of the control lever in the forward or upward direction shall cause forward motion of the truck.

3.24.7. Load Handling Controls. Load handling controls shall be self-centering levers located for convenient right hand operation within the maximum areas specified in SAE J898. Hoist and tilt controls may be combined in a single lever. Rearward or upward motion of a lever shall raise the forks, tilt the mast rearward, side shift the forks to the right, or position the forks closer together, as applicable.

3.24.8. Container Tophandler Controls. Tophandler controls shall be self-centering and grouped together for right hand operation and shall be located separate from load handling controls. Controls shall operate all tophandler operations including reaching, sideshift, retracting, slewing and twist lock activation. Twist lock activation may be kept separate from container top handling controls and may be arranged for either left or right hand operation.

3.25. Electrical System. The truck shall have a nominal 12 or 24 volt electrical system. Multiple conductor routing shall be protected by flexible tubing or plastic electrical insulation tape. Grommets shall be provided whenever wiring passes through bulkheads, partitions, or structural members; and the wiring shall be fastened to prevent chafing or abrasion. All wiring shall be color or number coded and the legend for such coding shall be included in the technical manual. All circuits except starting shall be furnished with circuit breakers in accordance with SAE J553 or fuses in accordance with SAE J554. All electrical connections fastened with terminal nuts shall be equipped with lock washers or nut locking devices. Circuit breakers shall be grouped in convenient locations and suitably marked for size and function.

3.25.1. Lights. Each truck shall be equipped with a minimum of eight sealed beam floodlamps or halogen lamps. The floodlamp

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sealed beam units shall conform to SAE J598, which shall be shock mounted in elastomer ring housings. Each light shall be adjustable a minimum of 45 degrees above and below horizontal plane. Two lights shall be adjustable to illuminate a minimum of 180 degrees to the front of the truck to provide forward illumination for driving when the truck is carrying a container in the transport position. Two lights shall be adjustable to illuminate 180 degrees to the rear of the truck. Two lights shall be adjustable in order to illuminate two twistlocks and of such design they are aimed at the twistlocks. Two lights shall be located such as to illuminate the rear of the container ie, the longitudinal side nearest the operator. Adequate protection against damage of each light, by means of position or guard, shall be provided.

3.25.2. Tophandler coupler indicator lights. A minimum of two tophandler coupler position indicator lights located in the cab adjacent to tophandler controls shall be furnished. A green or white light shall indicate all twist locks are down in the container corner fittings for both the locked and unlocked position. A red or amber light shall indicate all twist locks are in the locked position. Any one corner not properly or fully engaged will prevent the remaining lights from functioning until full engagement is proven. Indicator lights shall be provided with a marking diagram or plate indicating each coupler.

3.25.3. Taillight. One automotive type combination stoplight and taillight shall be furnished. The taillight shall conform to SAE designations as specified in SAE J759. The lamps shall be either sealed units mounted in elastomer rings or bulbs and lenses mounted in elastomer housings. If furnished, the bulb shall be Trade No. 1157. The taillight shall conform to SAE ST1, red, minimum 3 inch (75 mm) diameter, and shall be recessed or mounted on the rear and within the plan outline of the truck, and shall be protected against damage by a guard.

3.25.4 Horn. The truck shall be equipped with an electric horn with a pushbutton mounted in the center of the steering wheel or foot operated within reach of operator's seated position. The horn button assembly and electrical wiring for the horn shall be constructed to be moisture and weather-resistant to prevent entry of moisture when operated or stored outdoors under all weather conditions.

3.25.5. Direction Indicators. Electric automotive type turn indicators shall be mounted on both sides of the front and rear of the truck visable from the road level in accordance with standard automotive practice to indicate turning direction. A control lever shall be mounted on the steering column within reach of the seated truck operator. A signal type device visable

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to the operator shall indicate the turn indicators operation.

3.26. Operator's Overhead Guard. An overhead guard shall be furnished. It shall be capable of withstanding the impacts as specified in ASME/ANSI B56.1 without fracture or permanent deflection exceeding the allowable limits. The overhead guard shall be constructed in a manner that does not interfere with operator visibility, but openings in the top shall not exceed 6 inches (150 mm) in one of the two dimensions, width or length. The guard shall extend over the operator under all normal circumstances of truck operation. Overhead guard height shall be a minimum of 39 inches (990 mm) above seat reference point as specified in SAE J898 or shall be of sufficient height and construction to allow installation of the cab specified herein under the guard.

3.27. Personnel Cab. The truck shall be equipped with a weathertight and thermal and sound insulated weathertight personnel cab. Cab clearances shall be in accordance with SAE J154. The cab shall inherently resist torsional strain and withstand all operational shock and vibration under the operating conditions specified herein. The cab shall be constructed of metal. The cab shall be equipped with a minimum of two fully opening or removable doors or one fully opening or removable door and one emergency exit. Each door shall have a safety latch, opening stop, and a hold open catch. All hardware shall be the industrial type and resistant to corrosion. The cab shall be equipped with a minimum of two ventilation fans of not less than 6 inches (150 mm) in diameter. Fans shall be directionally adjustable to direct air flow toward the door and rear window glazing. Each fan shall have an "on-off" switch. In lieu of two ventilation fans, one ventilation fan may be furnished when the cab design permits the ventilation fan to direct air flow as specified herein.

3.27.1. Glazing. The cab shall provide the operator with all around visibility. The total glazed area shall be not less than 4000 square inches (2.6 square meter) of tinted windows consisting of windshield, sliding or roll-down windows, on both sides of cab, rear window, roof window, and other windows as determined by cab design. All glazing shall conform to ANSI Z26.1, Type AS1 in the windshield and roof window and Type AS1 or AS2 in other windows. Fixed glazing shall be sealed in waterproof stress absorbent molding and moveable windows shall be sealed on extra heavy type, pile lined channels. The roof window does not have to be furnished if the operator's line of sight is unobstructed when viewed from the operator's normal operating position to the two top corner container couplings nearest the operator when containers are stacked to the specified stacking height (see 6.2) when tophandler is engaging the container.

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3.27.2. Windshield and Window Wipers. The cab shall be equipped with at least two powered wiper assemblies to clear the windshield and rear window, and one powered wiper assembly to clear the roof window, when furnished. Windshield wiper systems shall be in accordance with the requirements of SAE J198. Wiper frequency shall be fixed or variable speed. Variable stroke wipers shall be capable of speeds up to 60 cycles per minute minimum. Fixed speed wipers shall be single speed capable of 60 cycles per minute minimum. Self-parking features shall be provided. Wiper arms and blades shall be of the commercial heavy-duty type, having saddle and pin blade fastening. The wiper arms shall provide tension on the glazed surface of at least 1-ounce (.28 N) for each inch (25 mm) of wiper arm length. Air motors shall be plumbed to exhaust air outside of the cab. The operator controls shall be located for convenient operation by the operator form the normal seated operating position.

3.27.3. Interior Lighting. A dome light shall be furnished and controlled by a separate switch. The switch shall be located for operation by the seated operator. The dome light shall be of the truck van type having a lens with an area of not less than 12 square inches (7740 square mm). The dome light shall be located in a position to provide reading illumination at laplevel.

3.27.4. Personnel Cab Heater and Defroster. Unless otherwise specified (see 6.2), a air type heater and defroster that utilizes engine coolant for heating shall be furnished. Heater shall be of sufficent capacity to maintain a temperature of 50° F (10°C). at cab floor level when the truck is operating at the specified ambient temperature (see 6.2). Heater shall be equipped with a removeable safety guard.

3.28. Safety Features. Each truck shall be equipped with the following safety features:

3.28.1. Overload Alarm. An audible warning device shall be located in the cab and shall sound an alarm to warn the operator of an overload condition when the tophandler is attached to a container and the container gross weight is one percent in excess of rated load. A silencing switch is not allowed.

3.28.2. Rearview Mirror. The truck shall be equipped with an adjustable rearview mirror mounted within the personnel cab.

3.28.3. Sounding Device. A directional sounding device, which operates continuously in each direction of travel shall be furnished. The sounding devices shall be activated for both the forward and rearward directions. Sounding device shall conform to SAE J994.

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3.28.4. Lights. Flashing amber lights visable from all directions, which operate continuously when the truck is in operation. The lights shall conform to SAE J96 or SAE J845. A minimum of three lights shall be provided, two on the front of the truck and one on the rear. The lights shall be mounted between three to six feet (915 to 1830 mm) above ground level and shall not interfere with truck operations or be exposed to damage from loading and unloading operations. One flashing amber light may be furnished in lieu of three lights provided it is visable within a 24 foot (7.3 m) diameter circle.

3.29. Treatment and Painting. The truck shall be painted contractor"s standard commmerical yellow paint except neither lead or chromate primer or finish coat paint is permitted. All exterior surfaces of the truck shall be thoroughly cleaned and shall be dry and free from mill scale, oil, grease, dirt, and rust and shall be painted, as soon as practicable after cleaning, with not less than one coat of base primer and not less than one coat of synthetic enamel or lacquer. Primer and finish coats shall be applied in accordance with normal commercial practices to assure complete coverage and durability of the finish. The finish coat, when dry, shall be a smooth even surface, free from runs, sags, peels, chips, blisters, areas of thin film or no film. Surfaces of components and assemblies of the truck not normally painted with a finish coat shall be cleaned and prepared in accordance with standard commercial practice for the industrial truck industry. Lift chains, sprockets, rollers, plastics, fabric hoses (except hydraulic hoses) and drivebelts shall not be painted, except that slight overspray will be accepted provided there is no interference with the functioning of the component. Paint shall be kept off working surfaces where interference with working parts would result. Engine accessories, i.e., alternator, starter, radiator core and instruments, plated or painted by the manufacturer of such items, shall not be painted by the contractor and shall be protected or masked from overspray.

3.29.1. Walkway Coating. Floor plates, personnel cab floor and step surfaces shall be coated with a non-slip deck covering compound or shall be expanded metal capable of supporting a 250-pound (115 kg) man.

3.29.2. Salt Spray Protection. When salt spray protection is required (see 6.2), the finishes indicated in paragraphs 3.29.2.1 thru 3.29.2.4 shall also be provided in addition to the contractors standard commerical finish as specified herein (3.29).

3.29.2.1. Subassemblies. All purchased subassemblies and components shall be thoroughly cleaned and shall be free from

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mill scale, oil, grease, dirt and rust. Then apply a red primer compatiable with the following epoxy primers to all surfaces.

3.29.2.2. Fabrications. All fabrications manufactured by the contractor shall be shot blasted to clean all surfaces of mill scale, oil, grease, dirt and rust. Then apply a two component epoxy resin red primer to a 50 micron minimum dry film thickness. On both purchased subassemblies and contractor fabrications multiple coats are permisssable to acheive dry film thickness.

3.29.2.3. Primer Painting. After the truck and tophandler are assembled the entire truck shall be cleaned and a two component gray epoxy primer shall be applied to a minimum dry film thickness of 100 microns. Multiple coats are permissable to acheive dry film thickness. Allow truck to dry thoroughly and then apply a two pack epoxy micaciouse iron oxide primer to a minimum dry film thickness of 150 microns.

3.29.2.4. Final Painting. Again allow truck to dry thoroughly and apply one coat of contractor's standard commerical yellow air drying enamel. Finally, apply multiple coats of wet on wet air drying enamel to a minimum dry film thickness of 250 microns.

3.30 Identification Plates. The truck shall be equipped with identification plates conforming to MIL-P-514 as specified herein. All plates shall be securely attached to the truck with screws, bolts, or rivets, and shall be furnished and mounted by the contractor.

3.30.1 Identification marking plate. Each truck shall be identified with an identification plate conforming to MIL-P-514, type I, style 1, Composition C, Grade A, Class I, which shall be mounted in a visible and safe location on the instrument panel or fire wall of the truck. The identifying data to be applied to the identification plates shall be as indicated herein and as defined in MIL-STD-130. Identification data shall include the truck lifting capacity, lift height, name of manufacturer, model number, serial number, contract number, gross vehicle weight (without rated load), registration number, delivery date, technical manual stock number, shipping weight (without tophandler or forks), cube dimension, and name or stamp of government inspector. The tophandler and fork assembly shall be identified with an identification plate conforming to MIL-P-514, type I, style 1, Composition C, Grade A, Class I. Identification data shall include name of manufacturer, model number, serial number, contract number, gross weight, registration number, technical manual stock number, shipping weight, cube dimension, and name or stamp of government inspector.

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3.30.2. Instruction, warning, and caution plates. Where required, each truck shall be equipped with instruction, warning, and caution plates conforming to MIL-P-514, type III, composition C, grade A, class I, prominently located and describing any special or important procedure to be followed in operating, lifting, and servicing of the truck or its components. A load plate in accordance with shall be furnished in the operator's compartment, clearly visible from the normal operating position, in the vicintiy of the hydraulic controls with the following instruction:

"WARNING - DO NOT HANDLE LOADED (see 6.2 for required container length) FOOT CONTAINERS IN EXCESS OF (see 6.2 for required truck lifting capacity) POUNDS."

3.30.3. Shipping Data Plate. Shipping data plates conforming to MIL-P-514, Type III, and Composition C, Type I, Class 1 material shall be furnished for the truck, mast, counterweight, tophandler and fork assembly and shall show the silhouette of the truck, mast and tophandler in transport position indicating the location and capacity of the slinging and tiedown attachments and supplementary points of tiedown. The plate shall be attached by screws, bolts, or rivets in a conspicuous location. Wheel loading information specified in 3.32.2 may be combined in one plate with the requirements specified herein.

3.30.4. Wheel Loading Plate. Each truck shall be equipped with a wheel loading plate conforming to MIL-P-514, Type III, and Composition C, Type I, Class 1 material. The plate shall be attached by screws, bolts, or rivets in a conspicuous location. As a minimum the plate shall have the following information:

Wheel loading (no load)
Drive wheels (each) pounds (kg)
Steer wheels (each) pounds (kg)

Wheel loading (Gross vehicle weight)
Drive wheels (each) pounds (kg)
Steer wheels (each) pounds (kg)

3.30.5. Safety Rating Plate. Each truck shall be equipped with a safety designation plate conforming to MIL-P-514, Type III,and Composition C, Type I, Class I material. As a minimum, the plate shall indicate truck manufacturer's name and complete model number, safety designation (Type D or DS), independent testing laboratory's name and registration or index number assigned to the truck and date and location of inspection.

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3.31. Truck Marking. Markings on trucks shall be painted black enamel. Characters shall be block-type capitals and Arabic numerials. Location, information, and character size shall be as follows:

Information	Example	Location	Minimum Character Size (height)
Agency	U.S. Navy	On each side and rear	6 inches (150 mm)
Registration Number	USN 13-76151	On each side and rear	3 inches (75 mm)
Capacity (Tophandler)	60,000lbs (27,215 kg)	on each side	3 inches (75 mm)
Capacity (Forks)	60,000 lbs (27,215 kg)	On each side	3 inches (75 mm)
Safety	No smoking No riders	On rear of mast	3 inches (75 mm)
Tire pressure	TPXX	On frame above each tire	1 inch (25 mm)
Warning	Do not handle loaded 40 foot containers in excess of 67,200 (30,800 kg) or 72,800 lbs (33,020 kg)	On rear of tophandler visible to operator	3inches (75 mm)
Slinging/ tiedowns	Lift here	At slinging/tiedown	1 inch (25 mm)
Fuel	Diesel Only	Near fuel tank filter	1 inch (25 mm)
Paint marking	Painted with lead/chromate free paint by manufacturer (name), month and year	Each side of truck	1 inch (25 mm)

3.32. Workmanship.

3.32.1. Castings and Forgings. All parts, components, and assemblies of the forklift which include castings and forgings shall be clean of harmful extraneous material such as sand, dirt, sprues, scale and flux. Rework shall be limited to procedures which do not reduce mechanical properties or affect function. Castings with fastener holes shall provide for correct seating of fasterners such as bolt, lockwasher and nut assemblies and capscrews.

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3.32.2. Metal Fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame cutting, using tips suitable for the thickness of the steel may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly, except in the performance of designed heat treatment. External surfaces shall be free of burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.32.3. Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the forklift to conform to the operational requirements specified herein. Parts to be joined by filler welds shall be brought into as close contact as possible and in no event shall be separated by more than 3/16 inch (5 mm) unless appropriate bridging techniques are used. The welding process used in fabrication of the forklift shall be at the option of the contractor.

3.32.4. Bolted Connections. Boltholes shall be accurately formed and shall have the burrs removed. Washers, lockwashers or other positive locking devices shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement.

3.32.5. Riveted Connections. Rivets shall fill the holes completely. The upset rivet heads, shall be in full contact with the surface of the member, and shall be in accordance with SAE J492 or Industrial Fastener Institute Standard No. 114 for break mandrel blind rivets.

3.32.6. Machine Work. Tolerances and gages for metal fits shall conform to the limits specified herein, and to the standards of the materials handling equipment industry.

3.33. Performance.

3.33.1. Lifting Speed. Speed of lift with rated capacity load shall be not less than 30 feet per minute (.155 m/sec) over the entire distance from ground level to maximum lift height.

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3.33.2. Lowering Speed. Speed of lowering tophandler or forks with rated load shall be not more than 80 feet per minute (.405 m/sec) over the entire distance from maximum lift height to ground level. The hydraulic control system shall be damped, metered, or manually regulated to reduce shock and prevent truck from overturning when rated load is lowered at maximum speed and then stopped quickly. Speed of lowering unloaded tophandler or forks shall be not less than 40 feet per minute (.200 m/sec) over the entire distance from maximum lift height to ground level.

3.33.3. Turning Circle Diameter. The truck with tophandler fully retracted shall turn in a 54 feet (16,460 mm) maximum outside diameter circle for capacity codes "A" thru "D" and 70 feet (21,335 mm) for capacity code "E" and "F".

3.33.4 Tophandler connection time. The complete tophandler shall be capable of being connected to the truck and capable of full operation as specified herein and disconnected from the truck in a maximum of 1/2 hour.

3.33.5. Speed. While carrying the rated load on a level surface, truck shall be capable of attaining 15.0 mph (24.2 km/h) minimum speed in both the forward and reverse directions in high range speed.

3.33.6. Slope Ascension, First Speed, Forward Direction. Trucks shall be capable of ascending a 20 percent grade on a dry surface with and without rated capacity load. Trucks shall be able to accelerate from a dead stop on this slope, when carrying rated load.

3.33.7. Upright or Cantilever Boom Tilt. When unloaded, truck masts shall have minimum 2-1/2 degrees (plus 1/2 degree and minus 0 degrees) forward tilt and minimum 7 degrees (plus or minus 1/2 degree) rearward tilt from true vertical. Positive means shall be furnished to prevent cavitation of the tilt cylinders at minimum 120°F (49°C) oil temperature when tilting rated load to full forward tilt at engine low idle speed. Maximum time to tilt from full rear tilt to full forward tilt at rated height with rated load shall be 10 seconds at governed engine speed.

3.33.8. Collapsed Mast Height. Collapsed mast height of the unloaded truck measured from the ground to the highest point of the truck when tophandler and the fork assembly is in the lowest position shall not exceed the dimension specified (see 6.2).

3.33.9. Lift height (Tophandler Assembly). The truck minimum lift height (with rated load) shall be capable of stacking and unstacking the number of containers with the container height as

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specified herein (see 6.2). The truck shall be capable of engaging the specified containers when the container is a minimum of 4 inches (100 mm) below the ground level of wheels.

3.33.10. Lift height (Fork Assembly). The truck lift height with rated load and when equipped with the fork assembly shall not be less than that specifed herein (see 6.2).

3.33.11 Engagement.

3.33.11.1. Out of Level Container. The tophandler shall be capable of engaging and lifting the rated load when one end of the container is a minimum of 20 inches (505 mm) higher (40 foot container) or 10 inches (250 mm) higher (20 foot container) than the other end and the truck wheels are on level ground.

3.33.11.2. Slew. The tophandler shall be capable of engaging and lifting the horizontal rated load in the 20 through 40-foot container(s) when the length of the container is at a minimum of 3.0 degrees angle to the front of the truck carriage.

3.33.11.3. Reach. The tophandler shall be capable of engaging rated load in the 20- through 40-foot container at ground level when rated load is a minimum of 14 inches (355 mm) forward of the vertical uprights without movement of the truck. The tophandler shall be capable of depositing rated load in the 20-foot through 40-foot container on a minimum 8 foot 0 inches (2440 mm) high platform a minimum of 14 inches (355 mm) from the leading edge of the platform with the uprights vertical.

3.33.12. Underclearance. Minimum underclearance beneath the truck mast assembly, axle assemblies and frame with rated load shall be not less than 8 inches (205 mm).

3.33.13. Drift. The truck lift assembly shall be capable of holding the rated load at maximum lift height not less than 10 minutes with not more than 1.5 inches (40 mm) of vertical drift and not more than 1 degree of rotational drift from the vertical.

3.33.14. Stability. The truck shall meet the longitudinal and lateral stability requirements of the Federation Europeenne De La Manutention, Section IV, for Industrial Trucks (FEM Standard 4.001n).

3.33.15. Noise Limits. The noise level of the truck (excluding horns) shall not exceed 85 dB(A) at the operator's station and 3.28 feet (1000 mm) from truck exterior when tested in accordance with SAE J919.

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3.33.16. Surface Bearing Pressure. The surface bearing pressure of the truck with rated capacity load shall not exceed 110 p.s.i. (758 kPa) for capacity codes "A"thru "D" and 120 p.s.i. (827 kPa) for capacity code "E".

3.33.17. Overall Truck Width. The overall width of the truck with tophandler fully retracted shall not exceed 20 feet (6095 mm). The overall width of the truck without tophandler shall not exceed 150 inches (3810 mm) for capacity codes "A" thru "D" and 156 inches (3960 mm) for capacity code "E".

3.33.18. Axle Weight. The axles weight on the forklift truck axles with tophandler and with and without rated load shall not exceed the weights specified (see 6.2).

3.33.19 Truck Overload. The truck, related assemblies and forks shall show no permanent deformation or failure as the result of the 200 percent overload static test.

3.33.20. Service Brakes. Application of wheel brakes shall require not less than 35 pounds (16 kg) or more than 100 pounds (45 kg) of force. The truck with and without rated load shall be capable of stopping in both foward and reverse directions, at maximum speed specified herein, in a distance not greater than 100 feet (30,480 mm).

3.33.21. Parking Brake. The parking brake shall be capable of retaining the truck with and without rated load on a dry 20 percent grade.

3.33.22. Uneven Container Loading. Truck and tophandler assembly shall be capable of supporting 60 percent of the rated load by weight when the weight is distributed in 50 percent of the container length measured from one end without exceeding truck stability or permanent deformation or structural failure due to simulated uneven container loading.

3.33.23. Truck Operation. The truck shall complete 300 circuits. The test shall be conducted continuosly (24 hours per day) at a rate of 4 to 6 circuits per hour minimum until 300 circuits are completed. However, 300 circuits must be completed without any failures. If failure(s) occur before completion of 300 circuits, the failure must be corrected and test restarted beginning at the first circuit.

3.33.24. Electromagnetic Radiation. Unless otherwise specified (see 6.2), the truck shall not exceed the limits of electromagnetic radiation specified in SAE J551. When specified (see 6.2), the truck electromagnetic radiation shall be suppressed in accordance with Class C1, Group 2, part 8 of MIL-

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STD-461.

3.33.25. Steering. The steer wheels shall be capable of being turned from extreme right to extreme left or vice versa in not more than six complete turns of the hand steering wheel. The maximum steering effort shall be not more than 15 pounds (7 kg) rim pull in either direction when the unloaded truck is stationery on a smooth, dry concrete surface with the engine at low idle speed. The steering system shall be capable of withstanding the maximum attainable unbalanced tangential force of 150 pounds (70 kg) applied at the rim of the steering wheel without failure or permanent deformation.

3.33.26. Maintenance Operations. Unless otherwise specified herein, each operation in the following maintenance operations list shall be accomplished by one man in not more than the time specified, using common tools and the special tools, if any, furnished with the truck:

(a) Remove, replace, and adjust all engine-driven belts - 3/4 hour. Exclude engine timing belt(s).

(b) Remove and replace alternator - 2 men - 1/2 hour.

(c) Remove and replace all filters, screens, and strainers in all hydraulic systems - 1/2 hour.

(d) Drain engine coolant, remove and replace engine coolant system hoses and thermostat, and refill system - 1 1/2 hour.

(e) Drain engine lubricating oil, remove and replace oil filter elements,and refill crankcase - 1/2 hour.

(f) Remove and replace fuel filter elements - 1/4 hour.

(g) Disconnect battery cables, remove and replace batteries, and reconnect battery cables - 2 men - 1/2 hour.

(h) Drain torque converter oil and transmission oil remove and replace all filter elements and strainers, and refill converter and transmission - 1 hour.

(i) Remove and replace starter - 1 hour.

(j) Adjust brakes - 2 men - 3/4 hour.

(k) Disassemble and service air induction system including the cleaning of and replacement of the filter element - 1/2 hour.

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(1) Remove, read, and replace fluid level dipstick or conduct a visual inspection - 2 minutes each.

3.33.27. Tires. Tire loading, when the truck is carrying rated load (centered), shall not exceed the load limits specified in the T&RA Yearbook or European Tyre & Rim Technical organisation at 15 mph (24 km/h). In lieu of load ratings indicated in either tire organization , written verification and calculations of stating the permitted tire load ratings from the tire manufacturer is acceptable.

3.33.28. Visability. Forward visability shall be such as to permit the operator an unobstructed view to the level road from the drive tires to two truck lengths with container in the transport position. Mirrors, cab position or mast design can be used to meet this requirement.

3.33.29. Fork spacing. Fork spacing shall be simultaneously hydraulically adjustable from 36 inches (915 mm) maximum when together to 96 inches (2440 mm) minimum when spread. The dimensions shall be measured from the outer edges of the forks.

3.33.30. Fork sideshift. Fork sideshift capable of sideshifting the rated load a minimum of 12 inches (305 mm) each side of center (24 inches (610 mm) total travel).

3.33.31 Transportability. The container handler forklift truck shall be transportable by highway, rail, marine and air modes to the extent specified herein (see 6.2).

3.33.31.1 Air Transportability. The forklift truck shall be air transportable in a drive-on/drive-off mode ground level to aircraft and vice versa. The forklift shall meet all requirements and tests in MIL-STD-1791 and AFSC Design Handbook DH 1-11 applicable to transportation in the class(es) of aircraft as specified (see 6.2). Additionally, the forklift shall, with lubricating and hydraulic systems filled to capacity and the fuel tank at least 25 percent full, have a maximum per axle load not in excess of that permitted in AFSC Design Handbook DH 1-11 in the drive-on and drive-off mode.

3.33.31.2 Disassembly of Vehicle for Air Transportability. Components such as the counterweights, tophandler, cab, exhaust stack, mast, overhead guard, etc., may be removed from the forklift if such action is necessary to meet dimensional or weight requirements for aircraft drive-on loading. Using only a mechanic with common hand tools, both the removal and the reassembly of such parts shall not require longer than 4 hours maximum. Because of limited capacities of the tools in the field mechanic's tool box, all socket drives greater than 3/4 inch (20

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mm), and sockets and wrenches greater than 2-inch (50 mm) required for removal or reassembly of the parts shall be furnished with each forklift by the manufacturer. No accessories or attachments which would prevent the forklift from being driven on/off the aircraft via its own power shall be removed. The operators manual shall include complete instruction for omponent removal and replacement. The vehicle, after preparation for air transport, shall not exceed the allowable height and width dimensions listed in AFSC Design Handbook DH 1-11.

3.33.31.3 Highway Transportable. The forklift truck shall be highway transportable in CONUS by the M746/M747 Heavy Equipment Transporter without requiring certification as Essential to the National Defense. The exceptions to this requirement will be the Vermont permit limit or width and the District of Columbia permit limit for height. Limited disassembly without cutting or welding is acceptable for for meeting CONUS highway permits.

3.33.31.4 Marine Transportable. The forklift truck shall be transportable on the LCU 1466 and larger marine vessels in the US Army Lighterage Fleet.

3.33.31.5 Railroad Transportable. The forklift truck shall be rail transportable in accordance with the requirements of the Association of American Railroads (AAR) General Rules Governing the Loading of Commodities on Open Top Cars.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order,the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified (see 6.2) in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and a part of the contractor's overall inspection system or quality control program. However, the absence of any inspection requirements in the specification shall not relieve the contractor of the responsibliity of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it

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commit the Government to acceptance of defective material.

4.1.2 Acceptability criteria. Trucks that conform to all requirements in Section 3 and 5 of this specification and pass all applicable inspections and tests in Section 4 of this specification will be considered acceptable by the Government.

4.1.3 Component and material inspection. The supplier is responsible for assuring that components and materials used are manufactured, examined, and tested in accordance with the requirements of referenced specifications and standards.

4.2 Classification of inspection. Inspections and tests shall be classified as follows:

(a) First article (first produced truck) inspection and tests (see 3.1, 4.3 and 6.3).

(b) Quality conformance (for each production truck) inspection and tests (see 4.6).

(c) Inspection comparison (see 4.7).

(d) Inspection of preparation for delivery (see 4.8).

4.3 First article (first produced) truck inspection and tests. When a first article inspection is required (see 6.2), the truck(s) should be a first production item. The first article should consist of one unit except a second unit may be used for the air transportability test when it is required (see 6.2). When specified (see 6.2) the Government will conduct any or all of the first produced tests and inspections as listed in the ordering data. Any changes or deviations during production from the approved first produced truck shall be subject to the approval of the contracting officer. Approval of the first produced by the Government shall not relieve the contractor of his obligation to furnish truck(s) conforming to this specification. Prior to inspection and test of the truck(s) the following shall be performed:

(a) Furnish a schedule of maintenance to be followed during all testing of the first produced truck. A list of tools required to perform this maintenance shall also be provided.

(b) Service the truck with oils and greases specified herein and designated for use in the ambient temperature at which the tests will be conducted.

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show no evidence ... laps and seams. (See 3.1.7 and 3.3.3). The stud shall

(c) If required, break in the truck as prescribed by the supplier.

4.3.1 Inspections.

4.3.1.1 Pretest inspection. Prior to testing according to 4.3.2, one or more of the first produced trucks as specified in 3.1 and 6.2 shall be inspected for the defects marked "X" in Column 1 of Table V.

4.3.1.1.1 Safety inspection. The contractor shall include in the First Article Test report satisfactory evidence that the trucks furnished under this specification meet or exceed the requirements of UL558. Acceptable evidence of meeting these requirements shall be certified test reports from Underwriters Laboratories indicating that the trucks conform to the requirements of UL558.

4.3.2 Tests. Upon successful completion of the inspections specified in 4.3.1.1 and 4.3.1.1.1, the truck shall be subjected to the tests marked "X" in Column 1 of Table VI. Acceptance of a first produced truck shall not exclude the remaining trucks from the quality conformance inspection and test provisions specified herein. Test conditions and schedule shall be as follows:

(a) Unless otherwise specified herein, tests shall be conducted at the ambient temperature and climatic conditions existing at the place of test. The test surface, unless otherwise specified or specific tests shall be level within + 1 degree. Only that maintenance established by the supplier and submitted as a maintenance schedule prior to commencement of tests shall be performed during the tests.

(b) Tests shall be in accordance with Table VI and may be conducted in any order desired except Drift, Stopping Distance, and Parking Brake Tests shall be conducted after the Truck Operation Test. An "X" in the applicable column indicates the tests that shall be conducted.

(c) Rated load, container length, width, and height shall be as specified. When multiple container lengths are specified the longest container length shall be used for test loads unless specified otherwise in the test requirement. Container dimensions shall be in conformance with the requirements of ISO 668, MIL-C-52661, MIL-C-52788 or as specified.

(d) Rated load or lifting capacity is defined as the total or gross weight (content weight plus container weight). The tolerances for weight shall be plus two percent and minus one percent. The container tophandler is not to be included as part

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of the rated load.

(e) When the supplier conducts the First Article Tests specified herein (see 6.2), a failure report shall be prepared (see 6.2.2) within 24 hours describing any incident of equipment malfunction or failure during the conduct of the First Article Test.

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Table V. Inspection schedule

First Produced Inspection 1	Quality Conformance Inspection 2	Item No	Defect 3	Requirements Paragraph 4
x		101.	Safety	3.3
x		102.	Lubrication and fittings not as specified	3.5.1 to 3.5.1.1
x		103.	Screw threads not as specified	3.7
x		104.	Fungus and moisture resistance not as specified	3.8
x		105.	Engine not as specified	3.9
x		106.	Alternator not as specified	3.9.2.1
x		107.	Starting system not as specified	3.9.2.2
x		108.	Batteries not as specified	3.9.2.3
x		109.	Battery mounting not as specified	3.9.2.4
x		110.	Battery cable terminals not as specifed	3.9.2.5
x		111.	Fuel system not as specified	3.9.2.6
x		112.	Air cleaner not as specified	3.9.2.7
x		113.	Cooling system not as specified	3.9.2.8
x		114.	Fan not as specified	3.9.2.8.1
x		115.	Thermostat not as specified	3.9.2.8.2
x		116.	Radiator not as specified	3.9.2.8.3
x		117.	Coolant hoses not as specified	3.9.2.8.4
x		118.	Governor not furnished	3.9.2.9
x		119.	Exhaust system not as specified	3.9.2.10
x		120.	Oil filter not as specified	3.9.2.11
x		121.	Drive assembly not as specified	3.10
x		122.	Transmission not as specified	3.10.1
x		123.	Hydraulic system not as specified	3.11
x		124.	Hydraulic lines not as specified	3.11.1
x		125.	Quick-disconnect fittings not as specified	3.11.2

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Table V. Inspection schedule (continued)

First Produced Inspection 1	Quality Conformance Inspection 2	Item No	Defect 3	Requirements Paragraph 4
x		126.	Hydraulic pump(s) not as specified	3.11.3
x		127.	Hydraulic filters not as specified	3.11.4
x		128.	Hydraulic reservior not specified	3.11.5
x		129.	Relief valve not as specified	3.11.6
x		130.	Control valves not as specified	3.11.7
x		131.	Cylinders not as specified	3.11.8
x		132.	Hose reels not as specified	3.11.9
x		133.	Wiper/scraper not as specified	3.11.10.1
x		134.	Guards not as specifed	3.11.10.2
x		135.	Cold weather system aids not as specified	3.12
x		136.	Upright and carriage not specified	3.13
x		137.	Cantilever boom not as specified	3.13.1
x		138.	Container top handler not as specified	3.13.2 or 3.13.3
x		139.	Coupling devices not as specified	3.13.2.1 or 3.13.3.1
x		140.	Aligning arms not as specified	3.13.2.2 or 3.13.3.2
x		141.	Forks not as specified	3.13.4
x		142.	Fork positioner not as specified	3.13.4.1
x		143.	Fork sideshift not as specified	3.13.4.2
x		144.	Wheels not as specified	3.14.1
x		145.	Tires not as specified	3.14.2
x		146.	Steering not as specified	3.15
x		147.	Service brake system not as specifed	3.16.1
x		148.	Power brake system not as specifed	3.16.1
x		149.	Air brake system not as specified	3.16.2
x		150.	Air-over-hydraulic brake not as specified	3.16.3

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Table V. Inspection schedule (continued)

First Produced Inspection 1	Quality Conformance Inspection 2	Item No	Defect 3	Requirements Paragraph 4
x		151.	Straight hydraulic brake not as specified	3.16.4
x		152.	Parking brake not as specified	3.16.5
x		153.	Truck access not as specified	3.17
x		154.	Seat not as specified	3.18
x		155.	Seat belt not as specified	3.18.1
x		156.	Cowl or hood not as specified	3.19
x		157.	Wheel guards not as specified	3.20
x		158	Towing device not as specified	3.21
x		159.	Slinging and tiedown provisions not as specified	3.22
x		160.	Instrument panel not as specifed	3.23
x		161.	Electrical system monitor not as specified	3.23.1
x		162.	Engine oil pressure monitor not as specified specified	3.23.2
x		163.	Engine coolant temperature specified	3.23.3
x		164.	Transmission temperature monitor not as specified	3.23.4
x		165.	Hour meter not as specified	3.23.5
x		166.	Fuel gage not as specified	3.23.6
x		167.	Air pressure monitor not as specified	3.23.7
x		168.	Transmission pressure monitor not as specified	3.23.8
x		169.	Controls not as specified	3.24
x		170.	Light switches not as specified	3.24.1
x		171.	Engine run control not as specified	3.24.2
x		172.	Starting motor switch not as specified	3.24.3
x		173.	Braking & inching or declutch control not as specified	3.24.4

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Table V. Inspection schedule (continued)

First Produced Inspection 1	Quality Conformance Inspection 2	Item No	Defect 3	Requirements Paragraph 4
x		174.	Accelerator control not as specified	3.24.5
x		175.	Transmission control not as specified	3.24.6
x		176.	Load handling controls not as specified	3.24.7
x		177.	Container top handler controls not as specified	3.24.8
x		178.	Electrical system not as specified	3.25
x		179.	Lights not as specified	3.25.1
x		180.	Tophandler coupler indicator lights not as specified	3.25.2
x		181.	Taillight not as specified	3.25.3
x		182.	Horn not as specified	3.25.4
x		183.	Direction indicators not as specified	3.25.5
x		184.	Overhead guard not as specified	3.26
x		185.	Personnel cab not as specified	3.27
x		186.	Glazing not as specified	3.27.1
x		187.	Windshield and window wipers not as specified	3.27.2
x		188.	Interior lighting not as specified	3.27.3
x		189.	Cab heater and defroster not as specified	3.27.4
x		190.	Overload alarm not as specified	3.28.1
x		191.	Rearview mirror not as specified	3.28.2
x		192.	Sounding device not as specified	3.28.3
x		193.	Safety lighting not as specified	3.28.4
x	x	194.	Treatment and painting not as specified	3.29
x	x	195.	Walkway coating not as specified	3.29.1
x	x	196.	Salt spray protection not as specified	3.29.2 to 3.29.2.4

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Table V. Inspection schedule (continued)

First Produced Inspection 1	Quality Conformance Inspection 2	Item No	Defect 3	Requirements Paragraph 4
x		197.	Identification plates not as specified	3.30
x		198.	Identification marking not as specified	3.30.1
x		199.	Instruction, warning & caution plates not as specified	3.30.2
x		200.	Shipping data plate not as specified specified	3.30.3
x		201.	Wheel loading plate not as specified	3.30.4
x		202.	Safety rating plate not as specified	3.30.5
x	x	203.	Truck marking not as specified	3.31
x		203.	Castings and forgings not as specified	3.32.1
x		205.	Metal fabrication not as specified	3.32.2
x		206.	Welding not as specified	3.32.3
x		207.	Bolted connections not as specified	3.32.4
x		208.	Riveted connections not as specified	3.32.5
x		209.	Machine work not as specified	3.32.6

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Table VI. Test Schedule

First Produced	Quality Conform.	Test	Test Para.	Require. Para.	Test Method #
1	2	3	4	5	6
x		Lifting speed	4.3.2.1	3.33.1	1
x		Lowering speed	4.3.2.1	3.33.2	1
x		Turning circle diameter	4.3.2.2	3.33.3	2
x		Tophandler disconnect time	4.3.2.3	3.33.4	2
x		Speed	4.3.2.4	3.33.5	3
x	x	Slope ascension	4.3.2.5	3.33.6	4
x		Tilt	4.3.2.6	3.33.7	5
x		Collapsed mast height	4.3.2.7	3.33.8	6
x		Tophandler lift height	4.3.2.7	3.33.9	6
x		Fork lift height	4.3.2.7	3.33.10	6
x	x	Engagement	4.3.2.8	3.33.11.1 to 3.33.11.3	7
x		Underclearance	4.3.2.5	3.33.12	4
x		Drift	4.3.2.9	3.33.13	8
x		Stability	4.3.2.10	3.33.14	
x		Noise limits	4.3.2.11	3.33.15	9
x		Surface bearing press	4.3.2.12	3.33.16	10
x		Truck width	4.3.2.13	3.33.17	6
x		Axle weights	4.3.2.14	3.33.18	10
x		Truck & fork overload	4.3.2.15	3.33.19	11
x	x	Service brakes and stopping distance	4.3.2.16	3.33.20	3
x	x	Parking brake	4.3.2.5	3.33.21	4
x		Uneven container loading	4.3.2.17	3.33.22	12
		Truck operations	4.3.2.18	3.33.23	13
x		Electromagnetic radiation	4.3.2.19	3.33.24	
x		Maintenance operations	4.3.2.20	3.33.26	16
x		Tire loading	4.3.2.21	3.33.27	10
x		Visability	4.3.2.22	3.33.28	17
x		Fork spacing	4.3.2.23	3.33.29	
x		Fork sideshift	4.3.2.23	3.33.30	
x		Air transportability, axle & floor loading	4.3.2.24, 4.3.2.24.1, 4.3.2.24.2	3.33.31.1	18
		Disassembly	4.3.2.25	3.33.31.2	18
x		Rail Impact	4.3.2.26	3.33.31.5	
x		Operating temperature	4.3.2.27	3.2.1	14
x		Rain	4.3.2.28	3.2.2	15
x		Slinging and tiedown	4.3.2.29	3.22	

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4.3.2.1 Lift and lowering speeds. Test the truck in accordance with Test Method No. 1. Failure to conform with the requirements of 3.33.1 and 3.33.2 shall constitute failure of this test.

4.3.2.2 Turning circle diameter. Test the truck in accordance with Test Method No. 2. Failure to conform with the requirements of 3.33.3 shall constitute failure of this test.

4.3.2.3 Tophandler disconnect time. Test the truck in accordance with Test Method No. 2. Failure to conform with the requirements of 3.33.4 shall constitute failure of this test.

4.3.2.4 Speed. Test the truck in accordance with Test Method No. 3. Failure to conform with the requirements of 3.33.5 shall constitute failure of this test.

4.3.2.5 Slope ascension, parking brake, and underclearance. Test the truck in accordance with Test Method No. 4. Failure to conform with the requirements of 3.33.6, 3.33.12 and 3.33.21 shall constitute failure of this test.

4.3.2.6 Tilt. Test the truck in accordance with Test Method No. 5. Failure to conform with the requirements of 3.33.7 shall constitute failure of this test.

4.3.2.7 Tophandler and fork lift heights and collapsed mast height. Measure the truck in accordance with Test Method No. 6. Failure to conform with the requirements of 3.33.8, 3.33.9 and 3.33.10 shall constitute failure of this test.

4.3.2.8 Engagement. Test the truck in accordance with Test Method No. 7. Failure to conform with the requirements of 3.33.11.1, 3.33.11.2 and 3.33.11.3 shall constitute failure of this test.

4.3.2.9 Drift. Test the truck in accordance with Test Method No. 8. Failure to conform with the requirements of 3.33.13 shall constitute failure of this test.

4.3.2.10 Stability. Conduct "longitudinal stacking", "longitudinal travel", "lateral stacking" and "lateral travel" tilting platform tests in accordance with FEM Standard 4.001n. Failure to meet the requirements specified in FEM Standard 4.001n shall constitute failure of this test.

4.3.2.11 Noise limits. Test the truck in accordance with Test Method No. 9. Failure to conform with the requirements of 3.33.15 shall constitute failure of this test.

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4.3.2.12 Surface bearing pressure. Test the truck in accordance with Test Method No. 10. Failure to conform with the requirements of 3.33.16 shall constitute failure of this test.

4.3.2.13 Truck width. Measure the truck in accordance with Test Method No. 6. Failure to conform with the requirements of 3.33.17 shall constitute failure of this test.

4.3.2.14 Axle weights. Weigh and record actual truck axle weights in accordance with Test Method No. 10. Nonconformance to the requirements specified herein (see 6.2) shall constitute failure of this test.

4.3.2.15 Truck and fork overload. Test the truck in accordance with Test Method No. 11. Failure to conform with the requirements of 3.33.19 shall constitute failure of this test. Failure is defined as any evidence of deformation, cracks, fractures, broken welds and hydraulic system leaks or ruptures. Test shall be deleted when conventional fork operation is not required (see 6.2).

4.3.2.16 Service brakes and stopping distance. Test the truck in accordance with Test Method No. 3. Failure to conform with the requirements of 3.33.20 shall constitute failure of this test.

4.3.2.17 Uneven container loading. Test the truck in accordance with Test method No. 12. Failure to conform with the requirements of 3.33.22 shall constitute failure of this test.

4.3.2.18 Truck operations. Test the truck in accordance with Test Method No. 13. Failure to conform with the requirements of 3.33.23 shall constitute failure of this test. Failure for the purpose of this test is defined as any malfunction which cannot be remedied by adjustment, repair or replacement action by organizational maintenance personnel using organizational tools and parts within 10 minutes or the totaled time of malfunctions exceeds 2 percent of the time required to complete 300 circuits and which may cause:

(a) Failure to commence operation, cessation of operation or degradation of performance below designated levels.

(b) Serious damage to truck by continued operation.

(c) Serious personnel safety hazard.

(d) Only routine maintenance as prescribed by the technical manual furnished with the truck is allowed. Normal maintenance is not considered a failure.

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(e) The truck shall show no signs of unusual or detrimental wear.

(f) Refueling and normal maintenance indicated in the technical manual is not considered a malfunction.

4.3.2.19 Electromagnetic radiation. Test the truck in accordance with SAE J551. When specified (see 6.2), test the truck in accordance with MIL-STD-462. As a minimum, the electromagnetic radiation test report shall include results of all testing and a list of test equipment and their locations used in the tests. The report shall also list all modifications made to the truck which were required to bring the truck within SAE recommended levels. Nonconformance to 3.33.24 shall constitute failure of this test.

4.3.2.20 Maintenance operations. Test the truck in accordance with Test Method No. 16. Failure to conform with the requirements of 3.33.26 shall constitute failure of this test.

4.3.2.21 Tire loading. Test the truck in accordance with Test Method No. 10. Failure to conform with the requirements of 3.33.27 shall constitute failure of this test.

4.3.2.22 Visability. Test the truck in accordance with Test Method No. 17. Failure to conform with the requirements of 3.33.28 shall constitute failure of this test.

4.3.2.23 Fork positioner and sideshift. Operate the tophandler and forks (when furnished, see 6.2) sideshift for 100 cycles at full speed with rated load from extreme right to extreme left. During the last 50 cycles the control shall be released abruptly when the load is approximately at midtravel and then the sideshifter shall be reactivated. Operate the sideshift without rated load to demonstrate sideshift total travel. Also, failure to complete 100 cycles of sideshift shall constitute non compliance. Test may be temporarily terminated to permit the hydraulic fluid temperature to decrease to the truck's normal operating temperature. Operate the fork positioner without rated load to demonstrate fork spacing. Nonconformance with the requirements of 3.33.29 and 3.33.30 shall constitute failure of this test.

4.3.2.24 Air Transportability. When specified (see 6.2), the supplier shall demonstrate in accordance with Test Method No. 18 that the vehicle meets all air transportability requirements in 3.33.31.1. Certification that these requirements and tests have been previously met will be acceptable in lieu of actual testing. At the contractor's option and cost, air transportability may be tested on a contractor supplied mock-up of the specified aircraft

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(see 6.2). In lieu of actual testing (drive-on/drive-off or mock up demonstrations), and when specified in the contract or order, the contractor may prepare a preliminary and final Transportation Report (see 6.2.2). Failure to conform to any requirements listed in MIL-STD-1791 and AFSC Design Handbook DH 1-11 shall constitute failure of this test.

4.3.2.24.1 Air transportable axle loads. Test the truck in accordance with Test Method No. 19. Failure to conform with the requirements of 3.33.31.1 shall constitute failure of this test. Axle loads in excess of those permitted in AFSC Design Handbook DH 1-11 shall be cause for failure of this test.

4.3.2.24.2 Floor Load. Test the truck in accordance with Test Method No. 2. Failure to conform with the requirements of 3.33.31.1 shall constitute failure of this test. Tire loads in excess of AFSC Design Handbook DH 1-11 shall be cause for rejection.

4.3.2.25 Disassembly of Vehicle for Air Transportability. When specified (see 6.2), test the truck in accordance with Test Method No. 18. Failure to conform with the requirements of 3.33.31.2 shall constitute failure of this test.

4.3.2.26 Rail Impact. When specified (see 6.2), test the truck in accordance with the Association of American Railroads (AAR), Section 1, Part 3 governing the loading of Department of Defense material on open top cars when loaded on a standard gear flatcar. Failure is defined as any permanent deformation to the truck or the inability to perform lifting or lowering the rated load, driving the truck in forward and reverse gear, operating the sideshift and fork positioner, and steering operations. Failure to conform with the requirements of 3.33.31.5 shall constitute failure of this test.

4.3.2.27 Operating temperature. When specified (see 6.2), test the truck in accordance with Test Method No. 14. Failure to conform with the requirement of 3.2.1 or as specified otherwise (see 6.2) shall constitute failure of this test. Failure is defined as one of the following:

(a) Engine does not start within 5 minutes after the first cranking.

(b) Engine does not run smoothly within 15 minutes after starting without unnatural or continued control manipulation and without use of starting aids.

(c) Instruments and gages do not operate.

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(d) Battery charging system does not show positive charge.

(e) Transmission cannot be engaged.

(f) Brakes, cylinders or controls will not operate.

4.3.2.28 Rain. Test the truck in accordance with Test Method No. 15. Failure to conform with the requirements of 3.2.2 shall constitute failure of this test.

4.3.2.29 Slinging and tiedowns. Test the truck in accordance with the test requirements specified in MIL-STD-209 and AFSC Design Handbook DH 1-11 when air transportability is specified (see 6.2). Failure to conform to the requirements of 3.22 and 3.33.31.1 shall constitute failure of this test.

4.4 First Article (first produced) inspection and test report. When specified in the contract or order, a First Article Inspection Report (4.3.1 and 4.3.2) shall be prepared. Each inspection, test result and cover shall be witnessed and verified by a government representative and shall bear the government representative's stamp.

4.5 Issuing first produced truck. The approved first produced truck used for inspections and testing shall be presented for acceptance as the last unit for delivery under the contract, including option quantities, after being rehabilitated to a new appearance. This shall include repainting, performing maintenance related items indicated in 3.6.1 and the furnishing of new tires.

4.6 Quality conformance inspection and tests. Prior to shipment of truck(s)and after approval of First Article Test Report each truck shall be inspected and tested as specified herein.

4.6.1 Inspections. Prior to testing according to 4.6.2 each truck shall be inspected for defects marked "x" in column 2 of Table V. Presence of one or more defects shall be cause for rejection.

4.6.2 Tests. Upon sucessful completion of inspections specified in 4.6.1, each truck shall be subjected to the tests marked "x" in column 2 of Table VI. Failure of one or more tests shall be cause for rejection.

4.6.3 Certification. Each Material and Inspection Report (DD Form 250) shall bear the Government Representative's stamp indicating the truck(s) have successfully passed inspection and testing requirements specified herein.

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4.7 Inspection comparison. The Government may select trucks at any time during the contract production period and subject the trucks to the inspections specified in Column 1 of Table V and to the tests specified in Column 1 of Table VI to determine conformance to the requirements of this specification at no cost to the contractor. The inspection will be performed by the Government, at a site selected by the Government, on units selected at random from those which have been accepted by the Government and will not include the previously inspected and tested truck(s). In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison, and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison. Contractor will be permitted to witness Inspection comparison.

4.7.1 Inspection failure. Failure of an inspection comparison truck to meet any requirement specified herein during and as a result of the inspection and tests specified in 4.6 shall be cause for rejection of the inspection comparison truck(s) and shall be cause for refusal by the Government to continue acceptance of production trucks until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no cost to the Government on trucks previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all trucks accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

4.8 Inspection of preparation for delivery. The preservation, packaging, packing,and marking shall be examined to determine compliance with MIL-STD-162.

5. PACKAGING

5.1 Preservation, packing, and marking. Each complete truck shall be preserved, packed and marked in accordance with MIL-STD-162 for type I mobile or type II crated, as specified (see 6.2). The degree of preservation and packing shall be level A or level B or level C as specified (see 6.2 and 6.4).

6. NOTES

(This section contains information of a general nature that may be helpful, but is not mandatory.)

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6.1 Intended use. Trucks described herein are intended for stacking, unstacking, and moving cargo containers in and around military depots and sea ports. Trucks are intended for use on prepared surface such as reinforced concrete. Due to the weight these trucks exert on surfaces it is advised the adequacy of the surface bearing strength be evaluated and confirmed prior to ordering truck.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.

(b) Truck classification codes: Specify lifting capacity, container length, container height, container width and container stacking codes (see 1.2).

(c) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

(d) When First Article Inspection is required, number of test trucks to be furnished for testing, time frame required for submission of First Article test report and location of test facilities (see 3.1, 4.1, 4.3, 4.4 and 6.3).

(e) When operating temperatures should be other than specified (see 3.2.1, 3.9.2.3, 3.9.2.8, 3.12 and 3.27.4).

(f) When salt spray and/or dust protection is required (see 3.2.4, 3.9.2.7, 3.9.2.8.3, 3.11.4, 3.11.5, 3.11.8, 3.11.10 thru 3.11.10.2 and 3.29 thru 3.29.2.4 and 6.5).

(g) When other than type "D" safety rating is required (see 3.3).

(h) When battery condition should be wet and charged (see 3.9.2.3).

(i) When the truck shall be equipped with either a fixed type container tophandler(s) (see 3.13.4) or a telescoping type to handle SIXCON or QUADCON type containers (see 3.13.2). Specify ISO or military container length and width. Specify either SIXCON or QUADCON type containers are to be handled.

(j) When conventional fork operation (forks, fork positioner and sideshift) is not required (see 3.13.3 thru 3.13.3.2).

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(k) When personnel cab heater and defroster are not required (see 3.27.4).

(l) The required truck collapsed mast height (see 3.33.8).

(m) The required truck conventional fork lift height (see 3.33.10).

(n) The required truck axle weights (see 3.33.18).

(o) When electromagnetic interference shall conform to MIL-STD-461 in lieu of SAE J551 (see 3.33.24 and 4.3.2.19).

(p) Transportability requirements (see 3.33.31 thru 3.33.31.5). Specify class(es) of aircraft (C-130, C-141 and/or C-5) and location of drive on/off test facility.

(q) When government will conduct tests and to what extent of testing (see 4.3 and 6.3).

(r) When air transportability and rail impact tests are required (see 4.3.2.24 thru 4.3.2.24.2, 4.3.2.25 and 4.3.2.26).

(s) When operating temperature (cold start) test is required (see 4.3.2.27).

(t) The required type, degree and level of preservation, packaging and packing (see 5.1 and 6.4).

(u) The Government costs per day when Government personnel and equipment are to be utilized in reassembling each truck(s). Also, specify level of security clearance required for contractor personnel supervising the reassembly operation (see 6.6).

(v) When training for Government personnel is required , specify the number and type of personnel (maintenance and operators) to receive training, classroom hours of instruction, number of sessions and level of security clearance for contractor training personnel (see 6.7). Also, specify if contractor's training manuals and visual aids are to be furnished to the government.

6.2.2 Data Requirements. When this specification is used in an acquisition and data is required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the

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contract or purchase requirements. Deliverable data required by this specification are cited in the following paragraphs:

Paragraph No.	Data Requirement Title	Applicable DID No.
4.3.2 (e)	Report, Failure	DI-R -21598
4.3.2.24	Transportability Report	DI-PACK-80880

6.3 First Article. The contracting officer should include specific instructions in acquistion documents regarding arrangements for examinations, approval of first article test results and disposition of first article(s) as specified herein. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Conditions for Use of Level B Preservation. When Level B preservation is specified (see 5.1 and 6.2), this level of protection should be limited to truck(s) which are transported over the seas onboard marine vessels and when long term storage is required.

6.5 Salt Spray Protection Warranty. When salt spray protection is specified (see 6.2), the contracting officer should include warranty clauses such that truck(s) shall be warranted against damage from ocean salt water spray and atmosphere for a period of seven years. Damage is defined as a truck requiring repainting due the effect of salt spray or atmosphere.

6.6 Reassembly. The contracting officer shall include in the solicitation the requirement that the manufacturer shall provide services for on-site supervision, personnel and equipment necessary to disassemble, unpackage, and reassemble equipment items whose physical size prevents shipment in a fully assembled condition. When specified (see 6.2), the Government will provide the necessary equipment and personnel required to reassemble the truck under the contractor's direct supervision. Contractor will reimburse the government for the cost of activity equipment and personnel. The contractor shall commence assembly operations within two weeks of being notified the truck has arrived at the activity. The government will accept delivery of the equipment when written confirmation by contractor's on-site representative indicates the truck is ready for safe operation. The contracting officer shall include in the solicitation the following information regarding reassembly requirements:

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6.6.1 Government Reimbursement Data. When government equipment and personnel are utilized (see 6.2), the following information and data shall be provided to the contractor in the ordering data:

6.6.1.1 Labor Rates. List activity labor rates for government personnel , per person, per day.

6.6.1.2 Equipment and Operators Rates. List activity total costs per day for both the operator and the equipment for the following equipment:

(a) 30 ton crane
(b) 20,000 pound forklift truck
(c) 40 foot tractor/trailer
(d) 40 foot tractor/ lowboy trailer
(e) Other (specify)

6.7 Training. When specified (see 6.2), the contracting officer shall include in the solicitation training requirements such that the manufacturer is required to provide the services of two persons (minimum) to provide on-site training for both operational and maintenance type personnel in both the operation and maintenance of the truck. The number and type of people (operator, maintenance,etc.) to receive training, the required number of training sessions, and time training hours will be conducted each day shall be provided to the contractor in the ordering data (see 6.2). Training shall commence one day after the contractor has certified the truck is ready for operation. Training shall include all aspects of truck operation, maintenance, and safety. Technical manuals furnished with the truck and contractor's standard training manuals and visual aids (if available) shall be used for the training. When specified (see 6.2), two copies of contractor's training manual and one copy of the visual aids shall be furnished to the Government for future training of Government personnel. The level of security clearance required for the contractor's training personnel shall be indicated in the ordering data (see 6.2).

6.8 Subject Term (Key Word Listing).

Telescoping Tophandler
Cargo Containers
Truck Lift
Diesel Engine Driven
ISO Containers
Military Containers

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Custodians:
Army ME
Navy SA
Air Force 84

Review Activities

Army - SM, MT
DLA - CS
Navy - OS, YD1, SH, MC

Preparing Activ

Navy SA

(Project 3930-0670)

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APPENDIX

FOR

TRUCK, LIFT, DIESEL ENGINE DRIVEN,

PNEUMATIC RUBBER TIRES WITH TELESCOPING TOPHANDLER

FOR HANDLING CARGO CONTAINERS

10. SCOPE The test methods contained within this appendix are to determine whether trucks procured under this specification conform to the requirements set forth.

20. APPLICABLE DOCUMENTS. None.

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TEST METHOD NO. 1

LIFTING SPEED - LOWERING SPEED

1. TEST COURSE:

(a) Level,flat surface.

2. TEST APPARATUS:

(a) Tape measure.
(b) Stop watch
(c) Cargo container with rated load (see 6.2)

3. LIFTING AND LOWERING TEST PROCEDURES:

a) Measure and record distance from ground to the bottom of the tophandler when the tophandler is in the horizontal position. Record time in seconds required to raise container with rated load from ground level to maximum lift with mast height in true vertical position.

(b) Record time to lower container with rated load at maximum speed (full open valve) to 3 to 4 foot (915 to 1220 mm) height.

c) Repeat steps 3(a) and (b) a minimum of three times. Test results shall be the average of the three tests results.

(d) Repeat step 3(b) without rated load.

(e) Repeat steps 3(a) to (d) when conventional fork operation is required (see 6.2).

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TEST METHOD NO. 2

TURNING CIRCLE DIAMETER AND

TOPHANDLER DISCONNECT TIME

1. TEST COURSE:

(a) Level, flat surface

2. TEST APPARATUS:

(a) Tape measure
(b) Stopwatch
(c) Cargo container with rated load (see 6.2)

3. TOPHANDLER DISCONNECT TIME TEST PROCEDURE:

(a) Record time to completely disconnect container tophandler from mast.

(b) Record time to completely connect tophandler to mast.

(c) Operate tophandler to demonstrate reach, slew and out of level container operations.

4. TURNING CIRCLE DIAMETER TEST PROCEDURE:

(a) Mark position of drive tires on test course.

(b) Negotiate a non-stop minimum turn of 360 degrees to the left in forward direction. Mark position of drive tires at 180 degrees.

(c) Measure and record distance between marks.

(d) Repeat steps 4(a) to (c) to the right in the forward direction.

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TEST METHOD NO.3

BRAKES - SPEED - STOPPING DISTANCE

1. TEST COURSE:
(a) Test course shall be a level surface of sufficient length to attain maximum speed prior to entering measured distance, plus measured distance, and sufficient stopping distance.

2. TEST APPARATUS:
(a) Tape measure.
(b) Stopwatch or electronic timer
(c) Dynamometer
(d) Cargo container with rated load (see 6.2)

3. SPEED TEST PROCEDURE:
(a) Operate the truck at governed speed with cargo container and rated load in carry position.

(b) Drive the truck a sufficient distance to attain maximum speed in high gear prior to entering the measured test course.

(c) Measure and record length of measured distance, time to traverse measured distance and engine speed. Measured distance shall be equal to or greater than 44 feet (13 m).

(d) Repeat for a total of six runs, three in forward gear and three in reverse direction. Test results shall be the average of the three test runs in each direction.

4. BRAKING FORCE AND STOPPING DISTANCE TEST PROCEDURES:

(a) Attach a pedal force gage to brake pedal in a manner which enables a force to be applied to the brake pedal.

(b) Repeat steps 3(a) through (c).

(c) At the end of the measured distance the operator shall apply the service brakes and safely bring the truck to a complete stop.

(d) A passenger shall record the minimum and maximum pedal force required to bring the truck to a complete stop.

(e) Measure and record the distance from the end of the measured distance to the front of the tophandler.

(f) Perform steps 4 (a) through (e) with rated load.

Note: Stopping distance shall be conducted after Test Method No. 14 has been successfully completed.

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TEST METHOD NO.4

SLOPE ASCENSION - PARKING BRAKE - UNDERCLEARANCE

1. TEST COURSE:

(a) Tilt table or ramp.

2. TEST APPARATUS:

(a) Clinometer.
(b) Spring scale.
(c) Cargo container with rated load (see 6.2)
(d) Underclearance template (optional)

3. SLOPE ASCENSION, PARKING BRAKE AND UNDERCLEARANCE TEST PROCEDURES:

(a) Drive truck forward with container and rated load along horizontal approach and onto ramp of specified percent grade and bring to complete stop. All wheels shall be positioned on the ramp.

(b) Apply parking brake.

(c) Release service brakes and observe whether truck remains stationary. Release parking brake.

(d) Starting from a standstill, proceed up ramp and position dddrive wheels onto a horizontal surface. Measure and record underclearance.

(e) From a horizontal surface drive truck in reverse, down slope, stop on slope, and repeat steps 3(b) through (e).

(f) Repeat steps 3(a) and e) without container.

(g) In lieu of driving onto a horizontal surface at top of incline, a template may be used to check underclearance.

(h) Measure and record underclearance beneath the mast assembly, beneath each axle and the frame of the truck. Measure and record the underclearance of the lowest component and note the component.

Note: Parking Brake Test shall be conducted after the Truck Operation Test (Test Method No. 13).

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TEST METHOD NO. 5

UPRIGHT OR CANTILEVER BOOM TILT - TILT SPEED

1. TEST COURSE

(a) Level, flat surface.

2. TEST APPARATUS:

(a) Clinometer
(b) Cargo container with rated load (see 6.2)

3. UPRIGHT OR CANTILEVER BOOM TILT TEST PROCEDURE:

(a) Place truck without container and rated load on a flat level surface, using clinometer to insure truck is level and mast is vertical.

(b) Raise tophandler to an elevation of 10 feet (3050 mm) above ground and tilt upright or cantilever boom as far forward as possible and record angle of tilt using clinometer located on front surface of upright or boom. Then tilt upright or boom as far rearward as possible and record angle of tilt on clinometer.

4. TILT SPEED TEST PROCEDURE:

(a) Anchor rear of truck by external means.

(b) Raise container with rated load to a minimum of 108 inch (2745 mm) height measured to the bottom of the container and then position the container at maximum rear tilt.

(c) Without adjustment of the height control tilt the container with rated load to full forward at maximum speed and record time.

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TEST METHOD NO. 6

COLLAPSED ~~MAST~~ HEIGHT - LIFT HEIGHT - OVERALL TRUCK WIDTH

1. TEST COURSE:

(a) Level, flat surface.

2. TEST APPARATUS:

(a) Tape measure.
(b) Plumb bob.
(c) Liquid level or clinometer.
(d) Container with rated load (see 6.2)

3. COLLAPSED MAST HEIGHT TEST PROCEDURE:

(a) With the upright and tophandler in the lowest possible position and without rated load measure and record the true vertical distance from floor to uppermost projection of the upright assembly or cantilever boom in the true vertical position as determined by a clinometer. (This is the actual collapsed mast height.)

4. LIFT HEIGHT TEST PROCEDURES:

(a) Position container with rated load 4 inches (100 mm) below the level of the drive wheels.

(b) Engage container with rated load with tophandler and raise to maximum lift height. Adjust the tilt such that the container is horizontal with the ground. Measure and record the true vertical distance from the ground to the bottom of the tophandler (top of container).

(c) Disconnect the tophandler and install fork assembly.

(d) Lower rated load to 12 inches (305 mm), plus or minus 4 inches (100 mm), above ground level and position mast in true vertical position using clinometer. Attach a plumb bob to the theoretical intersection of the front surface and top surface of the forks and mark a reference point on the ground.

(e) Elevate the rated load to the maximum lift height and check the reference point. If necessary, reposition the mast so that the plumb bob is directly over the original reference point. Measure and record the true vertical distance from test course to the top surface of the forks.

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TEST METHOD NO. 6 CONTINUED

COLLAPSED MAST HEIGHT - LIFT HEIGHT - OVERALL TRUCK WIDTH

5. OVERALL TRUCK WIDTH TEST PROCEDURE:

(a) Measure and record the overall truck width with tophandler at the widest point on the truck

(b) Measure and record the overall truck width with fork assembly at the widest point on the truck

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TEST METHOD NO. 7

ENGAGEMENT

OUT OF LEVEL - SLEW - REACH - SIDESHIFT

1. TEST COURSE:

(a) Level, paved surface.

2. TEST APPARATUS:

(a) Protractor.
(b) Block(s) capable of supporting one of the ends of specified container(see 6.2).
(c) Minimum 8 foot (2440 mm) high platform.
(d) Spacer blocks. 20 inches (505 mm) high x 20 inches (505 mm) wide x 8 foot (2440 mm) long and 10 inches (255 mm) high x 10 inches (255 mm) wide x 8 foot (2440 mm) long.
(e) Cargo container with rated load (see 6.2).

3. OUT OF LEVEL TEST PROCEDURE:

(a) Place one end of a 40 foot container with rated load on a 20 inch (505 mm) spacer block. Disengage tophandler from container, raise to clear corner fittings and level the tophandler.

(b) Re-engage 40 foot container with tophandler and raise to a 3 foot (915 mm) height measured to bottom of container.

(c) Move spacer block to opposite end of container and deposit container on block. Repeat steps 3 (a) and (b).

4. SLEW TEST PROCEDURE:

(a) From one corner of a 40 foot container with rated load, draw a line on the pavement such that the line forms a minimum 3.5 degree angle with the length of the container.

(b) Aline the truck such that the front face of the carriage is parallel with the line.

(c) Engage the container without backing the truck.

(d) Repeat steps 4 (a), (b) and (c) from opposite corner.

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TEST METHOD NO. 7 CONTINUED

ENGAGEMENT

OUT OF LEVEL - SLEW - REACH - SIDESHIFT

5. REACH TEST PROCEDURE:

(a) With 40 foot container with rated load at ground level position truck such that the truck vertical upright is 14 inches (355 mm) minimum from the container.

(b) Without any truck movement engage container and lift rated load to travel position.

(c) Truck shall proceed to the 8 foot (2440 mm) high platform and deposit container a minimum of 14 inches (355 mm) from leading edge of platform.

6. SIDESHIFT TEST PROCEDURE:

(a) With 40 foot container with rated load at ground level draw a parallel line 12 inches (305 mm) minimum from both ends of the container.

(b) Engage the container without moving the container, raise to 12 inches (305 mm) above ground level and sideshift to the right. Lower container to the ground.

(c) Observe if the end of the container has reached the line.

(d) Repeat steps 3(a),(b) and (c) and sideshift the container to the left.

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TEST METHOD NO.8

DRIFT

1. TEST COURSE:

(a) Level, flat surface

2. TEST APPARATUS:

(a) Temperature thermocouples
(b) Clinometer
(c) Cargo container with rated load (see 6.2)

3. DRIFT TEST PROCEDURE:

(a) Install thermocouple in the hydraulic reservoir. Raise the hydraulic oil temperature by exercising the lift and tilt cylinders until the normal operating temperature is reached. Normal operating temperature shall be maintained throughout this test.

(b) Raise container with rated load to maximum lift height obtainable with the outer mast channel in true vertical position measured with the clinometer. Then mark a reference point on mast upper stage cylinder.

(c) Record hydraulic oil temperature.

(d) Hold rated load for specified time and measure and record downward drift, rotational drift, and hydraulic oil temperature.

Note: The Drift Test shall be conducted after the Truck Operation Test (Test Method No. 13).

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TEST METHOD NO. 9

NOISE

1. Test Site:

(a) Smooth, uniform plane that has open space free of reflecting surfaces within 49 feet 3 inches (15 m) of forklift truck and in accordance with SAE J919.

2. Instrumentation:

(a) Sound level meter or microphone with magnetic tape recorder.
(b) Acoustical calibrator.
(c) Windscreen.
(d) Anemometer.
(e) Speed indicator.
(f) Thermometer.
(g) Barometer

3. NOISE TEST PROCEDURE:

(a) Operator's Station. Measure the maximum steady state noise levels of the truck in accordance with the procedures of SAE J919; except the following modes of operation shall be used in lieu of the tests indicated in SAE J919.

(1) Lifting rated load at maximum lift speed.
(2) Engine stalled against converter.
(3) Engine stalled against hydraulic system relief valve.
(4) Governed engine speed, no load.
(5) Maximum acceleration while carrying rated load.

(a) Exterior: Measure the maximum steady state noise levels of the truck at 3.28 feet (1000 mm) in accordance with the procedures of SAE J919; except the following modes of operation shall be used in lieu of the tests indicated in SAE J919.

(1) Lifting rated load at maximum lift speed.
(2) Engine stalled against converter.
(3) Engine stalled against hydraulic system relief valve.
(4) Governed engine speed, no load.
(5) Maximum acceleration while carrying rated load.

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TEST METHOD NO. 10

TIRE LOADING - SURFACE BEARING PRESSURE - AXLE WEIGHTS

1. TEST COURSE:

(a) Level, flat surface.

2. TEST APPARATUS:

(a) Platform or axle scales
(b) Air pressure gauge -0 to 150 psi (0 to 1035 kPa)
(c) Clinometer
(d) Cargo container with rated load (see 6.2)

3. TIRE LOADING TEST PROCEDURE:

(a) Steer wheels (without rated load): Determine and record the weight supported by each steer wheel and the axle by driving each steer wheels onto a platform scale or jacking each wheel up with an axle scale. The upright shall be in true vertical position, the tophandler mounted on the truck and the carriage positioned at maximum right sideshift.

(b) Repeat step 3 (a) with tophandler positioned at maximum left sideshift.

(c) Drive wheels (without rated load): Determine and record the weight supported by each drive wheel without container on the tophandler by driving each set of dual drive wheels onto a platform scale or jacking one set of dual wheels up with an axle scale. The total weight on each dual tire should be divided by two. The upright shall be in true vertical position, the tophandler shall be mounted on the truck and the carriage positioned at maximum sideshift.

(d) Repeat step 3 (c) with carriage positioned at maximum right sideshift.

(e) Drive and steer wheels (with rated load): Repeat steps 3 (a), (b), (c) and (d) with container and rated load.

4. SURFACE BEARING PRESSURE TEST PROCEDURE:

(a) When determining the tire loading also determine the actual tire area in contact with the surface. Divide the actual tire load by the actual tire area to determine surface bearing pressure.

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TEST METHOD NO. 10 CONTINUED

TIRE LOADING - SURFACE BEARING PRESSURE - AXLE WEIGHTS

5. AXLE WEIGHTS TEST PROCEDURE:

(a) Steer axle. Determine and record the weight supported by the steer axle without container on the tophandler by driving the steer axle onto a platform scale.

(b) Repeat steps 5(a) with container and rated load.

(c) Drive axle. Determine and record the weight supported by the drive axle without container on the tophandler by driving the drive axle onto a platform scale.

(b) Repeat steps 5(c) with container and rated load.

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TEST METHOD NO. 11

TRUCK AND FORK OVERLOAD

1. TEST COURSE:

(a) Level, flat surface
(b) Anchor

2. TEST APPARATUS:

(a) Magnetic particle or dye penetrant capability

3. TEST PROCEDURE:

(a) Place truck on level surface and support it with blocks under the axles or frame to relieve load on tires. Position mast in true vertical position forks positioned in the normal carrying position. External means may be utilized to lock the tilt cylinder.

(b) Secure truck in a manner to compensate for the test loads to be placed on forks. Counterweight may be added to assist the holding force. Apply counterbalancing forces in the vertical direction.

(c) With crane or another truck, place first test load of the rated capacity in position such that the center of mass of the test load is on the truck's longitudinal centerline and at rated load center.

(d) Check entire truck, hydraulic system and holddown devices to determine whether the additional test load can be placed on forks.

(e) With crane or another truck, place second test load of the rated capacity in position such that the center of mass of the test load is on the truck's longitudinal centerline and at rated load center.

(f) When placing the overload on the forks, personnel should stand clear and crane or truck operator should stand by to remove test load if there is any evidence of immediate collapse or truck structural failure.

(g) If truck accepts the second test load without immediate failure, let overload test load stand on truck for 10 minutes.

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TEST METHOD NO. 11 CONTINUED

TRUCK AND FORK OVERLOAD

(h) Remove test loads and inspect truck structure, frame, mast assembly, hydraulic system for evidence of deformation, fracture, leakage in hydraulic system broken welds or any other failure. Record each failure.

(i) Inspect forks a minimum of 6 inches (150 mm) each side of heel by magnetic particle or dye penetrant for evidence of metal failure.

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TEST METHOD NO. 12

UNEVEN CONTAINER LOADING

1. TEST COURSE:

(a) Level, flat surface

2. TEST APPARATUS:

(a) Empty cargo container

3. TEST PROCEDURE:

(a) Place empty cargo container on a level surface.

(b) Load the container such that 60 percent of the rated load is placed in one half of the specified container length measured from one end. The other half of the container shall remain empty.

(c) Engage container with tophandler and raise to maximum lift height. Operate tilt mechanism as far forward and rearward as possible. Operate sideshift mechanism as far right and left as possible.

(d) During step 3 (c) observe truck if any wheel(s) have lifted of the ground. Obsevation of any truck wheel lifting off the ground indicates truck stability has been exceeded and constitutes failure of this test.

(e) Inspect the entire truck for deformation or structural failure.

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TEST METHOD NO. 13

TRUCK OPERATIONS

1. TEST COURSE:

(a) The test course shall be a level, improved surface laid out in accordance with Figure A-1. Station 1 and 3 shall be minimum 8 foot(2440 mm) high platforms capable of supporting a fully loaded cargo container as specified (see 6.2). Station 2 shall be ground level. At Station 2 place either a 20 inch (505 mm) or 10 inch (255 mm) spacer block at ground level such that one end of container will rest on block. Also, draw a line on the pavement such that the line forms a minimum 3.5 degree angle with the length of the container.

(b) Obstacles shall be in accordance with Figure A-2 and placed such that the nearest obstacle is not less than 10 feet (3050 mm) from Station 1.

2. TEST APPARATUS:

(a) Two 20 foot and two 40 foot cargo containers with rated load (see 6.2).

3. TEST PROCEDURE:

(a) Initally place rated loads on platforms 18 inches (455 mm) from leading edge at Stations 1 and 3.

(b) Pick up container from Station 1, back up truck being sure all wheels travel over obstacles, then travel forward and deposit load at Station 2 using the slew mechanism. One end of container shall be resting on the spacer block.

(c) After disengaging load at Station 2, backout, then travel forward to Station 3. Engage rated load at Station 3, raise mast to maximum lift height, operate sideshift in each direction, and operate forward and rearward tilt mechanism.

(d) Lower rated load to travel position, back out, and then travel forward and deposit load on platform 18 inches (455 mm) from leading edge at Station 1 being sure all wheels travel over obstacles.

(e) Disengage rated load, back out being sure all wheels travel over obstacles, and then travel to Station 2.

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TEST METHOD NO. 13 CONTINUED

(f) Engage load at Station 2, back out, then travel forward and deposit load 18 inches (455 mm) from leading edge of platform at Station 3.

(g) After disengaging load at Station 3, backout, then travel to Station 1.

(h) Steps 3(b) through (g) constitute one circuit. For 50 percent of the circuits, Station 2 and 3 shall be interchanged such that they are in clockwise position in lieu of initial counterwise.

(i) Record total circuits for each hour of operation.

(j) Record reasons for each stoppage and failures.

(k) Conduct half the test circuits with rated load in the 40 foot container and half the test with rated load in 20 foot container. If only a 20 foot container is specified (see 6.2) all tests will be conducted with a 20 foot container.

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FIGURE A-1 TRUCK OPERATIONS COURSE

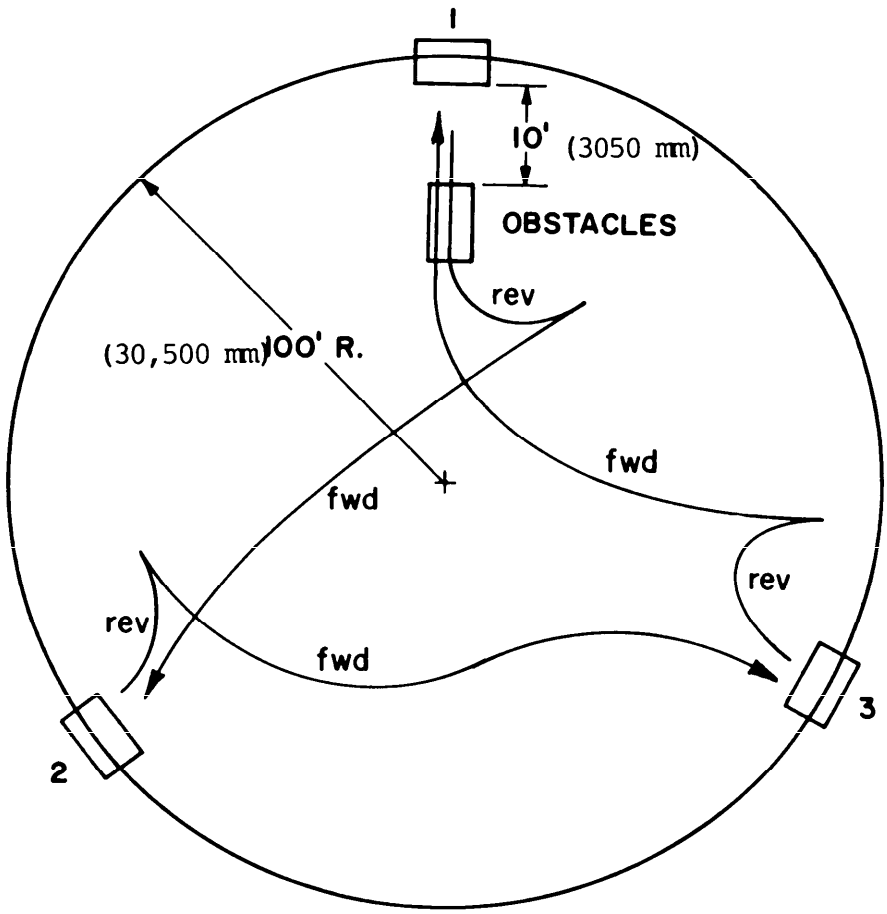
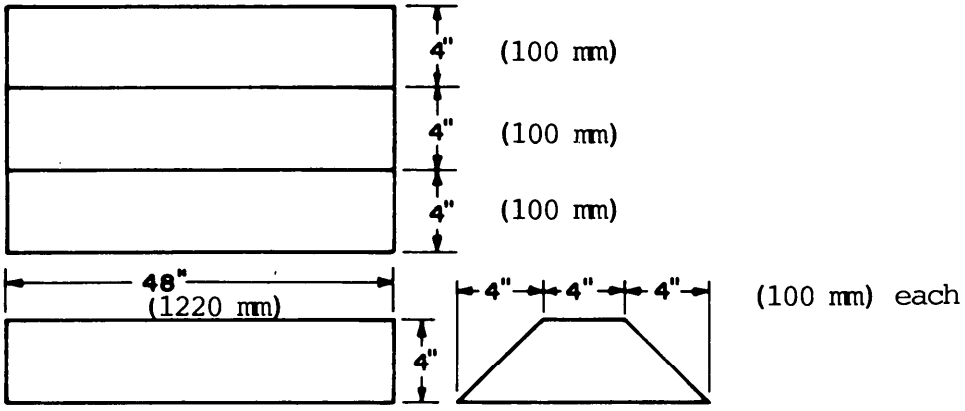


Figure A-1. ENDURANCE COURSE

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FIGURE A-2 DETAILS OF OBSTACLE CONSTRUCTION



LAYOUT OF OBSTACLE COURSE

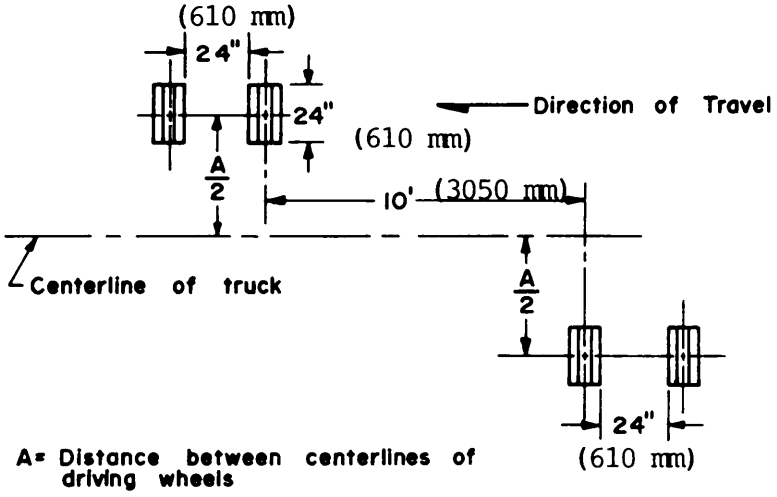


Figure A-2. DETAILS OF OBSTACLE CONSTRUCTION

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TEST METHOD NO. 14

OPERATING TEMPERATURE

1. TEST APPARATUS:

(a) Cold chamber
(b) Arctic clothing
(c) Thermocouples
(d) Battery hydrometer
(e) Wattmeter or ammeter
(f) Recording instruments (temperature and electrical)

2. TEST PROCEDURES:

(a) Service the truck with applicable lubricants for operation at temperatures as specified (see 6.2) including engine oil, transmission fluid, and drive train lubricant.

(b) Check the specific gravity of each battery to insure full electrical charge.

(c) Subject the truck to the specified ambient temperature until the truck, oils, lubricants,and coolant have stabilized at the specified temperature plus or minus 2° F (1°C). Perform the test after stabilization is assured by temperature readings from thermocouples located in the following locations:

(1) In the two center cells of each battery between the plates and at middepth of the electrolyte.

(2) In the volumetric center of the hydraulic system reservoir.

(3) In the engine lubricating oil gallery and in the oil sump.

(4) In the engine coolant within the engine cooling jacket discharge pipe.

(5) In the chamber at a horizontal angle of 45 degrees from the sides and end of the truck, and not less than three feet (915 mm) from each corner of the truck at engine crankshaft level.

(d) Maintain specified ambient temperature at plus or minus 2° F. (1°C) throughout the entire test.

(e) One operator wearing cold weather clothing, including mittens, shall perform all priming and starting conditions.

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TEST METHOD NO. 14 CONTINUED

(f) Record all thermocouple readings when the truck temperatures have stabilized to the ambient conditions.

(g) Record time required to start engine.

(h) Substitute or supplementary batteries or any other electrical cranking energy is not permitted.

(i) Auxiliary cold weather starting system accessories which are permanently mounted on the truck and normally used in starting the engine may be used. All starting aids required to start the engine shall be furnished on all production trucks.

(j) Smooth engine running without unnatural or continued controlled manipulation and without use of primers and priming fluids should be obtained within 15 minutes after commencement of cranking.

(k) After engine has started has run for 15 minutes, perform the following operations:

(1) 10 minutes idling
(2) 10 minutes half throttle
(3) 5 minutes full throttle

(l) Record temperatures prior to engine starting, immediately after engine starting, and at 10 minute cycles during engine warmup and cycling at the following locations:

(1) Battery
(2) Hydraulic tank
(3) Engine oil
(4) Engine coolant
(5) Cold chamber ambient air

(m) At the end of the engine 25 minute throttling and idling cycle the truck transmission , uprights, and brakes shall be operated such as to ensure the truck will be able to function at the specified operating temperature.

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TEST METHOD NO. 15

RAIN

1. TEST APPARATUS:

(a) Timing device
(b) Measuring device or rain gauge
(c) Spray fixture

2. TEST PROCEDURE:

(a) Subject truck to a simulated rainfall of not less than 4 inches (100 mm) per hour. The simulated rainfall shall impinge on the truck at an angle of 45 degrees plus or minus 5 degrees and impinge with a force equivalent to 30 mph (48 km/hr) wind. The simulated rainfall shall disperse uniformly over the surface of the truck.

(b) Expose the top and left side of the truck to the rain described above for a period of 30 minutes with engine off. With rain still falling. Start the engine and allow engine to idle for 15 minutes.

(c) Repeat 2(b) with top and front exposed.

(d) Repeat 2(b) with top and right exposed.

(e) Repeat 2(b) with top and rear exposed.

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TEST METHOD NO. 16

MAINTENANCE OPERATIONS

1. TEST FACILITY:

(a) The test facility shall consist of a shop area of sufficient size to permit unobstructed surveillance of the maintenance operations.

2. TEST APPARATUS:

(a) Common handtools
(b) Special tools if furnished with truck

3. TEST PROCEDURES:

(a) Examine truck for capability of adjustment, servicing or replacement of truck components.

(b) Demonstate the accessibility of drains and path of discharge of lubricants and coolants by activating or removing drainage controls.

(c) Perform, time and record each operation on the maintenance operations list specified herein.

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TEST METHOD NO. 17

VISABILITY

1. TEST COURSE:

(a) Level, flat surface

2. TEST APPARATUS:

(a) Four foot (1220 mm) high pole with stand
(b) Empty container

3. TEST PROCEDURE:

(a) Place truck on test course with operator in cab.

(b) Engage container and raise to transport position.

(c) Place the pole in front of the right front wheel and record visability.

(d) Repeat step 3(c) in front of left front wheel.

(e) Repeat the procedure in 3(c) and 3(d) moving the pole out from the truck 12 inches (305 mm) at each position described. Movement out from the front tires shall be at a 45 degree angle away from the truck. Repeat process until location of pole is two truck lengths away from truck.

(f) Confirm and record pole visability at all indicated locations whether sight is through direct sight or mirrors.

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TEST METHOD NO. 18

AIR TRANSPORTABILITY

1. TEST COURSE:

(a) Level, flat surface

2. TEST APPARATUS:

(a) Mock-up of the specified aircraft (Contactor's option in lieu of actual drive on-drive off test on the specified aircraft). Mockup is defined as a structure which reproduces the ramp angles, ramp crests, entrance and interior dimensions and the locations of the specified aircraft tiedowns.
(b) Platform or axle scale
(c) Common handtools

3. TEST PROCEDURE:

(a) Perform, time and record the time required to disassemble truck items such as counterweights, tophandler, cab, mast and overhead guard in order to meet the specified air transportable configuration.

(b) Determine and record the weight supported by the steer axle in the air transportable configuration by driving both steer wheels onto a platform scale or jacking up each wheel with an axle scale.

(c) Determine and record the weight supported by the drive axle in the air transportable configuration by driving both set of drive wheels onto a platform scale or jacking up each set of dual wheels with an axle scale.

(d) A government loadmaster will then drive the truck in the air transportable configuration onto and off the specified aircraft. The contractor will conduct all drive-on and drive-off tests when a mockup of the aircraft is utilized.

(e) Difficulties encountered in the loading operation shall be recorded with particular reference to interference with aircraft structure, damage to the cargo floor or when unusual positioning operations are required to stow the truck.

(f) The truck then shall be tied down in accordance with the restraint criteria specified in the Design Handbook DH1-11. Details on the arrangement of tiedown attachments to the aircraft floor shall be included in the test report.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2. 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-T-24960A	8 MARCH 1994

3. DOCUMENT TITLE: MILITARY SPECIFICATION, TRUCK, LIFT, DIESEL ENGINE, WITH TOPHANDLER

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

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NAVY SHIPS PARTS CONTROL CENTER	717 - 790 - 5003	430 - 5003
c. ADDRESS (Include Zip Code) CODE 0341 PO BOX 2020 , MECHANICSBURG, PA 17055-0788	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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